



Software Release Notes for the Packaging and Deployment Solutions

Release: UC-5.2.0
Date: May 2025

Affirmed Networks Inc.

A Microsoft Company

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Release Abstract

This document contains the software release notes for the Affirmed Networks UnityCloud current release.

This release has been produced in accordance with Microsoft's Secure Development Lifecycle, including processes for authorizing software changes, antimalware scanning, and scanning and mitigating security bugs and vulnerabilities.

New Features and Enhancements

This release contains the following new Helm Deployment pipelines and updated features and enhancements in the pipelines and their corresponding repositories (INFRA-DEV, LIB-CODE, or SVC-DEV) and files. An updated ARM template content and sample files are shown in this section as well.

PADS MN-OPM and SMF/UPF Decoupling Support

Overview

PADS Decoupling is a new feature that deploys a local Docker registry on NetworkCloud NF clusters thus decoupling the dependency on OPM registry. Fed-infra_dreg (Infra-Dreg) is a new service deployed as a Helm chart and when running (and populated) becomes the Docker registry for SMF and/or UPF. The Helm chart museum is still part of OPM and the source for deployment of the charts.

Infra-Dreg is defined as a PaaS dependency of SMF/UPF and deployed along with the other dependent PaaS Helm charts.

Deployment Workflow

The steps below are performed by PADS pipeline for fresh install of SMF or UPF to include Infra-Dreg (5.1 to 5.2). The release is pushed to MN-OPM Infra-Dreg via ACR-to-Ucreg copy pipeline. User modifies snapshot to include new fed (infra-dreg) and then runs PADS CnfOperationsDeploy pipeline which is set to pull images from MN-OPM infra-dreg. PADS make an API call to OPM to do the install which includes deployment of Infra-Dreg. To populate the NF Infra-Dreg, the Ucreg-to-Ucreg pipeline is run and the "retag" phase is needed to repoint the installed feds to the new Infra-Dreg. This final step is done by running the CnflssuUpgrade pipeline where the repoPath will be set to NF Infra-Dreg. Note this is a one-time event for 5.1 to 5.2 migration.

The steps are summarized below:

- PADS retrieve all sensitive data from AT&T KV to be added to the snapshot
- Run ACR-to-Ucreg pipeline to populate helm charts/images to MN-OPM
- Run CnfOperationsDeploy pipeline with snapshot for PaaS+Infra-Dreg+SMF/UPF selecting MN-OPM as registry
- Upon completion of previous deploy, run Ucreg-to-Ucreg pipeline to populates images to NF Infra-Dreg
- Run CnflssuUpgrade pipeline with same snapshot selecting NFInfra-Dreg as registry

Note: snapshot must be modified to include fed-infra_dreg parameters as well as use \$repo-path variable for all charts (see release notes and User Guide).

ISSU Workflow

The steps below are performed by PADS pipeline for ISSU of SMF or UPF to include Infra-Dreg (5.1 to 5.2). The release is pushed to MN-OPM Infra-Dreg via ACR-to-Ucred copy pipeline. User modifies snapshot to include new fed (infra-dreg) and then runs PADS CnflssuUpgrade pipeline which is set to pull images from MN-OPM infra-dreg. PADS make an API call to OPM to do the ISSU which includes deployment of Infra-Dreg. To populate the NF Infra-Dreg, the Ucred-to-Ucred pipeline is run and the "retag" phase is needed to repoint the installed feds to the new Infra-Dreg. This final step is done by running the CnflssuUpgrade pipeline again where the repoPath will be set to NF Infra-Dreg. Note this is a one-time event for 5.1 to 5.2 migration.

The steps are summarized below:

- PADS retrieve all sensitive data from AT&T KV to be added to the snapshot
- Run ACR-to-Ucred pipeline to populate helm charts/images to MN-OPM
- Run CnflssuUpgrade pipeline with snapshot for PaaS+Infra-Dreg+SMF/UPF selecting MN-OPM as registry
- Upon completion of previous ISSU, run Ucred-to-Ucred pipeline to populate images to NF Infra-Dreg
- Run CnflssuUpgrade pipeline with same snapshot selecting NFInfra-Dreg as registry

Note: snapshot must be modified to include fed-infra_dreg parameters as well as use \$repo-path variable for all charts (see release notes and User Guide).

Snapshot Patching

Below is the current fed-infra_dreg snapshot snippet that MUST be added to the existing snapshot file for Infra-Dreg to be installed on the NF. Note that the "nfAndPaasInstanceId" is set to "012345678901234567890123". This value must also be added to the "nfAndPaasIds" section. See Infra-Dreg snapshot example/snippet below:

```
"nfAndPaasInstanceId"
{
  "nfAndPaasInstanceId": "012345678901234567890123",
  "releaseName": "fed-infra-dreg",
  "namespace": "fed-infra-dreg",
  "description": "",
  "chartName": "fed-infra_dreg:[CHART VERSION]",
  "values" : "{ \"global\": { \"defaultServiceAccountName\": \"unity-cloud-
serviceaccount\", \"overrides\": { \"postDeleteEnabled\": false, \"postInstallEnabled\": true, \"postRol
lbackEnabled\": true, \"postUpgradeEnabled\": true, \"preCleanupEnabled\": false, \"preDeleteEnabled\":
```

New Features and Enhancements

```
false, \"preInstallEnabled\": false, \"preUpgradeEnabled\": true}, \"registry\": {\"docker\": {\"imagePullSecret\": \"\", \"repoPath\": \"$repo-path\", \"upstreamRepoPath\": \"$upstream-repo-path\", \"infraDreg\": {\"enabled\": true, \"loadBalancerIP\": \"<infra-dreg-lb-ip>\", \"port\": 5005, \"registryUrl\": \"<ucreg-url>\", \"secretName\": \"infradreg-cert-manager\", \"storageClass\": \"csi-sc-cinderplugin\", \"storageSize\": \"200Gi\"}}, \"registry_docker_path\": \"$repo-path\", \"secretNames\": {\"infradreg\": {\"userProvided\": {\"tlsCertSecretName\": \"secrets/data/tls-cert\", \"tlsKeySecretName\": \"secrets/data/tls-key\"}}}, \"test\": {\"loggingEnabled\": false, \"testPodInfraDreg\": true}, \"vault\": {\"enabled\": false, \"hashicorpVault\": {\"authMountPath\": \"jwt\", \"enabled\": false, \"role\": \"reader\", \"serverUrl\": \"http://fed-hc-vault-test-svc-internal.fed-hc-vault-test:8200\", \"vaultNamespace\": \"\"}}, \"pod-cert_updater\": {\"affinity\": {}, \"containerdCertsPath\": \"/etc/containerd/certs.d\", \"fullNameOverride\": \"\", \"hostTrustPath\": \"/etc/pki/ca-trust/source/anchors/\", \"image\": {\"pullPolicy\": \"IfNotPresent\"}, \"labels\": {\"k8spspallowedusers\": \"exclude\", \"k8spspallowprivilegeescalationcontainer\": \"exclude\", \"k8spspreadonlyrootfilesystem\": \"exclude\", \"livenessProbe\": {\"enabled\": true, \"failureThreshold\": 3, \"initialDelaySeconds\": 10, \"periodSeconds\": 10, \"successThreshold\": 1, \"timeoutSeconds\": 1}, \"nameOverride\": \"\", \"nodeSelector\": {}, \"podAnnotations\": {}, \"priorityClassName\": \"uc-priority-1\", \"probePort\": 28080, \"rbacEnabled\": true, \"readinessProbe\": {\"enabled\": true, \"failureThreshold\": 3, \"initialDelaySeconds\": 10, \"periodSeconds\": 10, \"successThreshold\": 1, \"timeoutSeconds\": 1}, \"registryUrl\": \"\", \"resources\": {\"limits\": {\"cpu\": \"100m\", \"memory\": \"100Mi\"}}, \"runtimeClassName\": \"\", \"secretName\": \"infradreg-cert-manager\", \"securityContext\": {}, \"serviceAccount\": {\"annotations\": {}, \"create\": true, \"name\": \"\"}, \"startupProbe\": {\"enabled\": true, \"failureThreshold\": 30, \"periodSeconds\": 5}, \"tolerateMaster\": true, \"tolerations\": [], \"updateStrategy\": {\"type\": \"RollingUpdate\"}}, \"pod-infra_dreg\": {\"affinity\": {}, \"autoscaling\": {\"behavior\": {}, \"enabled\": false, \"maxReplicas\": 2, \"minReplicas\": 1, \"targetCPUUtilizationPercentage\": 60, \"targetMemoryUtilizationPercentage\": 60}, \"certManager\": {\"certDuration\": \"2160h\", \"enabled\": true, \"issuerName\": \"affirmed-clusterissuer\", \"organization\": \"affirmednetworks\", \"pkiAlgorithm\": \"RSA\", \"pkiKeySize\": \"4096\", \"renewBefore\": \"648h\", \"configData\": {\"health\": {\"storagedriver\": {\"enabled\": true, \"interval\": \"10s\", \"threshold\": 3}}, \"http\": {\"addr\": \"5000\", \"debug\": {\"addr\": \"5001\", \"prometheus\": {\"enabled\": false, \"path\": \"/metrics\"}}, \"headers\": {\"X-Content-Type-Options\": {\"nosniff\"}}, \"log\": {\"fields\": {\"service\": \"registry\"}}, \"storage\": {\"cache\": {\"blobdescriptor\": \"inmemory\"}}, \"version\": 0.1}, \"containerSecurityContext\": {\"allowPrivilegeEscalation\": false, \"capabilities\": {\"drop\": [\"ALL\"]}, \"enabled\": true, \"privileged\": false, \"readOnlyRootFilesystem\": true, \"runAsGroup\": 1000, \"runAsNonRoot\": true, \"runAsUser\": 1000, \"selinuxOptions\": {}, \"seccompProfile\": {\"type\": \"RuntimeDefault\"}}, \"deployment\": {}, \"extraEnvVars\": [], \"extraVolumeMounts\": [], \"extraVolumes\": [], \"garbageCollect\": {\"deleteUntagged\": true, \"enabled\": false, \"resources\": {}, \"schedule\": \"0 1 * * *\"}, \"image\": {\"pullPolicy\": \"IfNotPresent\"}, \"imagePullSecrets\": [{\"name\": \"acr-secret\"}], \"ingress\": {\"annotations\": {}, \"className\": \"nginx\", \"enabled\": false, \"hosts\": [\"chart-example.local\"], \"labels\": {}, \"path\": \"/\", \"tls\": [], \"initContainers\": [], \"metrics\": {\"enabled\": false, \"port\": 5001, \"prometheusRule\": {\"enabled\": false, \"labels\": {}, \"rules\": {}}, \"serviceMonitor\": {\"enabled\": false, \"labels\": {}}, \"nodeSelector\": {}, \"persistence\": {\"accessMode\": \"ReadWriteOnce\", \"enabled\": true, \"size\": \"100Gi\", \"storageClass\": \"csi-sc-cinderplugin\"}, \"podAnnotations\": {}, \"podDisruptionBudget\": {}, \"podLabels\": {}, \"priorityClassName\": \"uc-priority-1\", \"proxy\": {\"enabled\": false, \"password\": \"\", \"remoteurl\": \"https://registry-1.docker.io\", \"secretRef\": \"\", \"username\": \"\"}, \"registryUrl\": \"\", \"replicaCount\": 1, \"resources\": {}, \"secrets\": {\"hasSharedSecret\": \"\", \"httppasswd\": \"\", \"securityContext\": {\"enabled\": true, \"fsGroup\": 1000, \"fsGroupChangePolicy\": \"Always\", \"supplementalGroups\": [], \"sysctls\": [], \"service\": {\"annotations\": {}, \"labels\": {}, \"loadBalancerIP\": \"\", \"name\": \"ucregistry\", \"port\": 5005, \"type\": \"LoadBalancer\"}, \"serviceAccount\": {\"annotations\": {}, \"create\": false, \"name\": \"\", \"storage\": \"filesystem\", \"tlsSecretName\": \"infradreg-cert-manager\", \"tolerations\": [], \"updateStrategy\": {}, \"userProvided\": {\"enabled\": false}, \"userProvidedVault\": {\"enabled\": false}}}, \"type\": \"PAAS\", \"chartType\": \"AFFIRMED_PAAS\", \"status\": \"DEPLOYED\", \"order\": 0, \"tooltip\": null, \"updated\": true, \"versionsMatch\": false, \"groupName\": \"<group-name>\", \"skipDeployCharts\": null, \"sharedParamsVersion\": 0, \"overlayConfVersion\": 0,
```

New Features and Enhancements

```
"issuSkip": false,  
"helmOptions": null  
}
```

Note there are several site-specific parameters in the snapshot snippet that are parameterized (enclosed in angle brackets <>) and values set in the input json file via the pipeline. Those params are:

<ucreg-url> - The Infra-Dreg external URL from input json data (nfUcReg.svcFqdn)

<ucreg-ip> - The Infra-Dreg external load balancer IP address from input json data (nfUcReg.svcIp)

<group-name> - Indicating the snapshot file group name which is extracted from the passed-in snapshot file. The pipeline replaces this with the group name from other feds in the snapshot file.

\$upstream-repo-path – The repo path for fed-infra-dreg registry pod to pull from. A daemonset runs on all masters and workers that prepulls and caches this image from upstream to decouple from MN-OPM. During install or upgrade, the new version of this image will pull from upstream and once again, be cached to all other workers to handle registry switchover (to a different node).

[CHART VERSION] – The fed-infra-dreg version must be manually set in the snapshot (e.g, 0.0.0-68-main-int).

All instances of repoPath entries in the snapshot (for all feds) MUST be replaced with \$repo-path variable. This variable is replaced by the pipeline to point to either the MNOPM Infra-Dreg or NF Infra-Dreg.

Once the snapshot file has been patched, proceed with CnfOperation “deploy” or CnfIssu upgrade pipelines.

UCReg Copy Image

UnityCloud Registries are nothing more than docker registries running as a service on Kubernetes clusters. Three use-cases are supported by Unity cloud:

- SMF-UCReg
- UPF-UcReg
- MNOPM-UCReg

Once the Infra-Dreg has been deployed on a cluster, it is ready for use but first must be populated with docker images. This is the same for any docker registry – cannot pull an image from registry if it has not been previously pushed.

Two new pipelines have been created for pushing images to Infra-Dreg – one is to copy from ACR to the MNOPM-Infra-Dreg and the second is to copy from MNOPM Infra-Dreg to SMF Infra-Dreg or UPF-Infra-Dreg. The latter pipeline is generic called UcRegToUcRegImageCopy.

ACR to MNOPM-UcRegCopy Images

The `acrToMnOpmUcRegImageCopy` pipeline has a set of params set in a json file which are documented in detail in the PADS User Guide. This pipeline is responsible for populating the MNOPM Infra-Dreg from ACR after Infra-Dreg has been deployed on MN-OPM cluster.

MNOPM-UCReg to NF-UCReg Copy Images

Once the Infra-Dreg has been deployed on the NF cluster, the `MnOpmUcRegToNfUcRegImageCopy` pipelines are run to copy images from the MNOPM-Infra-Dreg registry to the SMF-Infra-Dreg (in SMF case) and/or UPF-Infra-Dreg (in UPF case). When the UCReg-to UcReg pipeline has successfully run, the NF cluster can be upgraded so all Helm charts pull from local UCReg. This new pipeline uses input params which include source Infra-Dreg and destination UCReg. The input params are discussed in detail in the Decoupling Guide.

CnfUpgradelssu/CnfOperationDeploy (MNOPM-UCreg)

The Decoupling feature's main function is to deploy `fed-infra_dreg` as part of SMF or UPF deployments. To do that, the `fed-infra_dreg` snapshot snippet (above) is added by the user to the SMF/UPF snapshot to be deployed. Using the 5.2 `CnfUpgradelssu` or `CnfOperationsDeploy` pipeline, SMF/UPF charts and all dependent PaaS components (including `fed-infra_dreg`) are upgraded/deployed. New parameters are added to the json input file and are described in the PADS User Guide. The new params include "useUcReg" which has options of `mnOpmUcReg|nfUcReg|None`. For the initial install, set this param to "mnOpmUcReg" which indicates the images for all deployed charts will be pulled from MNOPM Infra-Dreg. The new parameters are listed below:

<code>useUcReg</code>	Specifies which Infra-Dreg to pull images from upon deployment "mnOpmUcReg" or "nfUcReg" or "none"
<code>opmUcReg . svcFqdn</code>	Specifies the FQDN of the MNOP-Infra-Dreg where images will be pulled from on deploy.
<code>opmUcReg . svcIp</code>	Specifies the IP address of the MNOPM-Infra-Dreg service.
<code>nfUcReg . svcFqdn</code>	Specifies the FQDN of the NF-Infra-Dreg where images will be pulled from on deploy.
<code>nfUcReg . svcIp</code>	Specifies the IP address of the NF-Infra-Dreg service.

CnfUpgradelssu/CnfOperationDeploy (NF-UCreg)

Once the `CnfUpgradelssu` or `CnfOperationDeploy` (`useUcReg = "mnOpmUcReg"`) issu/deployment is complete, the NF-Infra-Dreg is functional and can be populated via the `MnOpmUcRegToNfUcRegImageCopy` pipeline (discussed above). After the Infra-Dreg is populated, the

CnfUpgradelssu/CnfOperationDeploy pipeline is run a second time but the “useUcReg” param is set to “nfUcReg” which will perform an issu/upgrade of all charts retagging all charts/pods to pull images from the local Infra-Dreg (NfUcReg).

FinalizeCluster Pipeline Input Changes

Once the The finalizedCluster pipeline is enhanced to validate connection to Infra-Dreg in addition to OPM-dreg. This change requires addition of parameters to the input json file.

<code>useUcReg</code>	Specifies which Infra-Dreg to pull images from upon deployment “mnOpmUcReg” or “none”
<code>opmUcReg.svcIp</code>	Specifies the IP address of the MNOPM-Infra-Dreg service.

MNOPM Deploy

In 5.2, MN-OPM will use the same Infra-Dreg infrastructure as the NFs meaning deployment of MN-OPM via the PADS pipelines will deploy fed-infra_dreg. New params have been added to the json input file and are described in the PADS User Guide. The prams include “ucRegUrl”, “ucRegIp”, as well as OPM VM cluster params. Initial deployment of MN-OPM via the HelmDeploy pipeline will deploy fed-opm_multi along with all PaaS components including fed-infra_dreg. The new params are listed below:

<code>acrRepo</code>	The acr domain name where the images and charts are upload
<code>ucregUrl</code>	The external load balancer IP address of the Infra-Dreg to be a service.
<code>uregIp</code>	The Infra-Dreg URL applied to infra-dreg is at deployment.
<code>clusterParameters.ingressVM.publicIpAddress</code>	Load balancer IP address for MN-OPM k8s services
<code>clusterParameters.ingressVM. adminPrivateKey.reference.keyVault.name</code>	Key vault name where MN-OPM secrets are sored
<code>clusterParameters.ingressVM. adminPrivateKey.reference.secretName</code>	MN-OPM M2M private key name
<code>clusterParameters.vms[]. ipAddress</code>	MN-OPM VM IP address
<code>clusterParameters.vms[].port</code>	VM SSH Port
<code>clusterParameters.vms[].type</code>	Type of Node (Master/Worker)
<code>clusterParameters.vms[]. adminUsername</code>	SSH username to communicate with MN-OPM VMs
<code>clusterParameters.vms[]. dataplane</code>	Set to true if the node (VM) must have a label nodetype=dataplane

The helmDeploy pipeline deploys mn-opm and requires the following values added to the fed-opm_multi-values.yaml file:

```
global:
  registry:
    docker:
      repoPath: tk5af.a4opacketcore.microsoft.com/rel_build_docker
  infraDreg:
    enabled: false
```

MNOPM Update

Once the MN-OPM deployment has completed, the Infra-Dreg is functional and can be populated using the acrToMNOpMUCRegImageCopy pipeline (discussed above). After the Infra-Dreg has been populated, the helmUpdate pipeline is run which upgrades all MN-OPM charts and PaaS and retags to pull images from the local MNOPM-UCReg.

Enabling cluster edge registry in Nexus cluster

To enable a local copy of container images in the Nexus K8s cluster, enable Azure Operator Service Manager (AOSM) cluster registry (CR). This can be done by updating the network function extension using running the below command after doing azure login using Service Principal,

```
az k8s-extension update --resource-group <RESOURCE_GROUP> --cluster-name
<CLUSTER_NAME> --cluster-type connectedClusters --name networkfunction-operator \
--config global.networkfunctionextension.enableClusterRegistry=true \
--config global.networkfunctionextension.enableEarlyLoading=false \
--config global.networkfunctionextension.clusterRegistry.storageClassName=<STORAGE_CLASS> \
--config global.networkfunctionextension.clusterRegistry.storageSize=<STORAGE_SIZE> \
--version <EXTENSION_VERSION>
```

For more details on the cluster registry and how it works, refer to the below documents.

[Manage the Azure Operator Service Manager cluster extension | Microsoft Learn](#)

[Get started with Azure Operator Service Manager cluster registry | Microsoft Learn](#)

PADS Support of NF Level Rollback on CNF ISSU failure in Nexus

This feature enhances the rollback capabilities in the AON PADS CNF ISSU pipeline in case of upgrade failures, ensuring both the failed Network Function Application (NfApp) and all previously completed NfApps are reverted to their initial version state.

- A new variable “enableRollbackOnISSUFailure” is introduced in the “SVC-DEV/config/variables/pipeline-template-variables.yml” file to enable/disable rollback on CNF ISSU failure and is set to false by default.
- When this flag is enabled, ensure that NF ARM template has the changes to enable rollback on failure at both the NF App level and the entire NF level.
- Following is the sample CNF ARM template to be placed in armTemplateFolder and to be configured as input before running ACR copy artifacts pipeline as part of designer flow.

```
{
  "$schema": "https://schema.management.azure.com/schemas/2019-04-01/deploymentTemplate.json#",
  "contentVersion": "1.0.0.0",
  "metadata": {
    "_generator": {
      "name": "bicep",
      "version": "0.16.1.55165",
      "templateHash": "1292792834379325573"
    }
  },
  "parameters": {
    "nfdv": {
      "type": "object"
    },
    "nfdvId": {
      "type": "string"
    },
    "smfValues": {
      "type": "object"
    },
    "siteSpecificValues": {
      "type": "object"
    },
    "secretValues": {
      "type": "secureObject"
    }
  },
}
```

```

    "nfviIdValue": {
      "type": "string"
    },
    "rollbackEnabled": {
      "type": "bool",
      "defaultValue": false
    },
    "config": {
      "type": "secureObject",
      "defaultValue": "[union(parameters('smfValues'),
parameters('siteSpecificValues'), parameters('secretValues'))]"
    }
  },
  "variables": {
    "copy": [
      {
        "name": "nfAppOverrides",
        "count": "[length(parameters('nfdv').helmPackageInformation)]",
        "input": {
          "name":
"[parameters('nfdv').helmPackageInformation[copyIndex('nfAppOverrides')].name]"
,
          "deployParametersMappingRuleProfile": {
            "helmMappingRuleProfile": {
              "releaseNamespace":
"[parameters('nfdv').helmPackageInformation[copyIndex('nfAppOverrides')].releas
eNamespace]",
              "releaseName":
"[parameters('nfdv').helmPackageInformation[copyIndex('nfAppOverrides')].releas
e]",
              "helmPackageVersion":
"[parameters('nfdv').helmPackageInformation[copyIndex('nfAppOverrides')].versio
n]",
              "options": {
                "installOptions": {
                  "atomic": false,
                  "wait": true,
                  "timeout": "80",
                  "testOptions": {
                    "enable": false,
                    "timeout": "15",
                    "rollbackOnTestFailure": false
                  }
                }
              }
            }
          }
        ]
      }
    ]
  }
}

```



```

        "upgradeOptions": {
            "atomic": true,
            "wait": true,
            "timeout": "120",
            "testOptions": {
                "enable": true,
                "timeout": "15",
                "rollbackOnTestFailure": true
            }
        }
    }
}
},
],
"nfConfigurationOverrides": {
    "nfConfiguration": {
        "rollbackEnabled": "[parameters('rollbackEnabled')]"
    },
    "roleOverrides": "[union(variables('nfAppOverrides'),
array(variables('nfConfigurationOverrides')))]"
},
"resources": [
    {
        "type": "Microsoft.HybridNetwork/networkFunctions",
        "apiVersion": "2023-09-01",
        "name": "[format('aasm_smf_test{0}', uniqueString(resourceGroup().id))]",
        "location": "[parameters('nfdv').location]",
        "identity": {
            "type": "SystemAssigned"
        },
        "properties": {
            "configurationType": "Secret",
            "copy": [
                {
                    "name": "roleOverrideValues",
                    "count": "[length(variables('roleOverrides'))]",
                    "input":
"[string(variables('roleOverrides')[copyIndex('roleOverrideValues')])]"
                }
            ],
            "networkFunctionDefinitionVersionResourceReference": {

```

```

        "id": "[parameters('nfdvId')]",
        "idType": "Open"
    },
    "nfviType": "AzureArcKubernetes",
    "nfviId": "[parameters('nfviIdValue')]",
    "allowSoftwareUpdate": true,
    "secretDeploymentValues": "[string(parameters('config'))]"
  }
}
]
}

```

- Refer “Sample CNF ARM template” section in the user guide for more details on NF ARM template content.
- **Logging and Monitoring:**
 - Check the value of enableRollbackOnISSUFailure in the NSDV bicep parameter logged during the 'Display NSDV bicep file' stage in the 'CNF Operator ISSU' pipeline.
 - View the rollback option passed as part of the Role override values in the NF payload in the Azure portal under "Azure Network Function Manager – Network Function".
 - To identify if the NF Apps rolled back as expected, use helm list and helm history commands.

Pads idempotent in OPM Automation pipeline

This feature enhances the idempotent nature of the OPM Automation pipeline. The pipeline make sure that it does not fail when the pipeline is re run even though the chart repo, location and deployment zone is already created in the MNOPN when the pipeline is run earlier.

No changes in the configuration file need to be done as part of the feature all the required changes are available as part of the python script which are there in the docker images and the **affirmed-nr-opm-automation-micro-pipeline.yml** file is modified to call the MNOPM Api to make the pipeline idepotent.

Parallel Configparser runs to speed up execution of finalize cluster pipeline

As part of the feature the script 2.finalize.sh script has been modified to make the config parser script to run on all the nodes of the cluster from sequential execution to parallel execution. By parallel execution of the script the pipeline takes only 9 min to run the config parser on all the nodes.

The script ensures that if the config parser is already on the node. It ignores running the config parser script on that node and move to next node to validate if the config parser is already run run.

By parallel execution of the config parser the pipeline would take 9 min to run the finalize script as compared to earlier running of the sequential execution which would take 32 min to run the finalize script which has 3 master and 4 worker nodes there by significantly improving the performance of the finalizek8 cluster pipeline.

HashiCorp Vault Integration with MNOPM

Overview

The integration of HashiCorp Vault with MNOPM (Multi node OPM) federation is a strategic initiative aimed at enhancing the security and management of sensitive information within the Unity Cloud solution from UC 5.2 release. This integration focuses on migrating sensitive data such as usernames, passwords, connection strings, and certificates from local storage to the HashiCorp Vault, ensuring secure and centralized management of secrets.

Key Features

- **Vault Configuration:** Helm specific value JSON allowing users to enable or disable the HashiCorp Vault, set the server URL, define roles, and specify authentication methods.
- **Sensitive Data Management:** All sensitive data used in the fed-opm_multi (multi-node) deployment, including credentials and certificates, will be stored in the HashiCorp Vault. This eliminates plaintext sensitive data from Helm values and replaces them with vault path references.
- **Automatic Secret Rotation:** OPM will manage the rotation of secrets configured in the Vault server automatically, reducing the need for manual intervention.
- **Exclusions:** Single Node OPM is excluded from this integration due to its lightweight nature and the impracticality of using Kube add-ons and a running K8 cluster.

Goals

- Enhance security by centralizing the management of sensitive information.
- Ensure compliance with security standards by automating secret rotation and management.

This integration is part of the broader effort to improve the security and efficiency of the Unity Cloud solution, providing a robust framework for managing sensitive information across federations.

Prerequisites

To ensure a smooth integration of HashiCorp Vault with OPM, the following pre-requisites must be met:

- **Vault Server Management:** Each customer will manage their own HashiCorp Vault server.
- **Centralized Vault Server:** A centralized (external) Vault server will be used, with multiple Kubernetes clusters interacting with it.
- **Authentication Method:** JWT authentication method with static keys is preferred.
- **Secret Setup:** OPM secret information must be set up in the Vault before deployment.
- **PKI Issuer:** Vault will act as a PKI Issuer, and cert-manager will use secret-less authentication to interact with it.
- **Vault Availability and Configuration:** The availability and configuration of the Vault server will be managed by the customer's HashiCorp Vault team.

Steps to Integrate

Once the prerequisites are met, follow these steps in each repository to integrate the HashiCorp Vault server.

Network cloud Deployment

LIB-CODE

1. Vault configurations can be enabled by keeping the HashiCorp server settings in the helm specific input json file

LIB-CODE/code/helm_operation/helm-specific/helm_specific_values-template.json

Ensure the following variables are included with existing variables:

- "enable-global-vault": "<CHANGE-ME>"

- "enable-hashicorp-vault": "<CHANGE-ME>"
- "hashicorp-vault-server-url": "<CHANGE-ME>"

HashiCorp Vault server setting variables

Variables	Description
<code>enable-global-vault</code>	To enable the global vault option for the charts. Default is false.
<code>enable-hashicorp-vault</code>	To enable the HashiCorp Vault to read the secrets from the vault, by default false
<code>hashicorp-vault-server-url</code>	HashiCorp Vault server URL, basically vault server endpoint. For example, <code>http://fed-hc-vault-test-svc-internal.fed-hc-vault-test:8200</code>
<code>role</code>	HashiCorp Vault role to access secrets, default is reader role (Optional)
<code>authMountPath</code>	HashiCorp Vault Auth mount path, default is jwt (Optional)
<code>vaultNamespace</code>	HashiCorp Vault namespace, default is empty (Optional)

2. Ensure these tokens are configured in the respective values YAML of the required federations, from which credentials need to be read from the HashiCorp Vault.

- `\LIB-CODE\code\helm_operation\Values\fed-opm_multi.yaml`
- `\LIB-CODE\code\helm_operation\Values\fed-paas_helpers.yaml`
- `\LIB-CODE\code\helm_operation\Values\fed-kube_addons.yaml`

Ensure specify the HashiCorp server setting parameters in the values YAML file as outlined below.

```
vault:
  # Global switch to enable / disable vault configuration
  enabled: $enable-global-vault
  # Any one type of vault should be enable at a time
  hashicorpVault:
    # Switch to enable / disable Hashicorp vault
    enabled: $enable-hashicorp-vault
    # Hashicorp Vault server URL
    serverUrl: $hashicorp-vault-server-url
    # Hashicorp Vault role to access secrets
    role: "reader"
    # Hashicorp Vault Auth mount path
    authMountPath: "jwt"
    # Hashicorp Vault namespace
    vaultNamespace: ""
```

Note: Role, authMountPath, and vaultNamespace have default values and are optional. If needed, define these tokens in the Helm-specific input JSON file and mention them in the values YAML files for federations requiring secrets from the HashiCorp Vault.

3. Ensure the below vault secret paths configured in the fed-opm_multi yaml file, under the global section of the chart.

```
secretNames:
  fedopmmulti:
    opmUsernameSecretName: secrets/data/opmUsernameSecretName
    opmPasswordSecretName: secrets/data/opmPasswordSecretName
    mongoUsernameSecretName: secrets/data/mongoUsernameSecretName
    mongoPasswordSecretName: secrets/data/mongoPasswordSecretName
    mongolnitUsernameSecretName: secrets/data/mongolnitUsernameSecretName
    mongolnitPasswordSecretName: secrets/data/mongolnitPasswordSecretName
    mongoConnectionStringSecretName: secrets/data/mongoConnectionStringSecretName
```

Note: As mentioned in the prerequisite section, the secrets should be placed into the vault at the same secret vault path in the HashiCorp Vault server for MNOPM.

4. Run the helm deploy pipeline for the MNOPM installation and the helm update pipeline for the MNOPM upgrade.

Nexus Deployment

There will be no changes regarding Nexus Deployment as it is outside the scope of the HashiCorp Vault Integration with MNOPM

Verify the HashiCorp Vault Integration

Below are the steps to verify whether the secrets are being read from the HashiCorp Vault after the successful deployment / upgrade of MNOPM federation. Execute the following command on the master node using the command line interface (CLI).

kc get secrets -n vnfmgr

The TYPE of the MNOPM secrets (opm-auth, mongo-auth, and mongo-init-auth) should be "Opaque".

NAME	TYPE	DATA	AGE
kubeconfig-def765-0123456789abcdef	Opaque	1	12d
m2m-private-keys	Opaque	0	12d
mongo-auth	Opaque	3	12d
mongo-bootstrap-auth-data	Opaque	1	12d
mongo-init-auth	Opaque	3	12d
opm-auth	Opaque	2	12d
opm-unitycloud-com-ingress-cert-manager	kubernetes.io/tls	3	12d
sh.helm.release.v1.fed-opm-multi.v1	helm.sh/release.v1	1	12d
users	Opaque	1	12d

Updated pipelines with SUDO privileges

The PADS pipelines detailed in this section have been updated to allow SSH access to virtual machines and non-root users to execute commands. As a prerequisite:

- Create a non-root user on the target VMs
- Configure an SSH public key for this non-root user to enable SSH key-based authentication.
- Grant the user SUDO permissions to execute commands that require privileges.
- Configure the username parameters defined in the following pipelines for the non root user.
- Ensure that SSH private key secret names and key vault references are appropriately set for the non-root user.

code/finalize_cluster/finalizeK8sCluster.json@LIB-CODE

- ClusterParameters.ingressVM.adminUsername - SSH username to communicate with the jump VM.
- ClusterParameters.ingressVM.adminPrivateKey.reference.keyVault.name - Azure Key-Vault name with SSH private-key to be used to get to the ingress-VM.
- clusterParameters.ingressVM.AdminPrivateKey.reference.secretName - Azure Key-Vault secret-name where the SSH private-key to be used to get to the ingress-VM is kept.
- ClusterParameters.vmM2MKey.reference.keyVault.name - Azure Key-Vault name with SSH private-key to be used from the ingress-VM to the individual VMS that forms the K8s cluster.
- ClusterParameters.vmM2MKey.reference.secretName – Azure Key-Vault secret-name where the SSH private-key to be used from the ingress-VM to the individual VMS that forms the K8s cluster is kept.
- clusterParameters.vms[].adminUsername – SSH username to communicate with VMs.
- opm.sshUsername – SSH username to communicate to OPM

- `opm.keyVault.name` – Azure Key-Vault name with OPM information
- `opm.keyVault.opmPrivateKeysecretName` – OPM secret name on the Azure Key-Vault for the private SSH key.
- `multiNodeOPM.opmMasterVMPrivateKeySecretName` – MNOPM secret name on the Azure Key-Vault for the private SSH key.

code/opm_automation/opmAutomation.json@LIB-CODE

- `opm.sshUsername` – SSH username to communicate to OPM
- `opm.keyVault.name` – Azure Key-Vault name with OPM information
- `opm.keyVault.opmPrivateKeysecretName` – OPM secret name on the Azure Key-Vault for the private SSH key.
- `multiNodeOPM.opmMasterVMPrivateKeySecretName` – MNOPM secret name on the Azure Key-Vault for the private SSH key.

code/acr_opm_image_copy/acrToOpmCopy.json@LIB-CODE

- `multiNodeOPM.opmMasterVMPrivateKeySecretName` – MNOPM secret name on the Azure Key-Vault for the private SSH key.
- `opm.sshUsername` – SSH username to communicate to OPM
- `keyVault.name` – Azure Key-Vault name with OPM information
- `keyVault.opmPrivateKeysecretName` – OPM secret name on the Azure Key-Vault for the private SSH key.
- `keyVault.opmPublicKeysecretName` – OPM secret name on the Azure Key-Vault for the public SSH key.

code/cnf_operations/cnfoperations-deploy.json@LIBCODE

- `multiNodeOPM.opmMasterVMPrivateKeySecretName` – MNOPM secret name on the Azure Key-Vault for the private SSH key.

code/cnf_operations/cnfoperations-delete.json@LIBCODE

- `multiNodeOPM.opmMasterVMPrivateKeySecretName` – MNOPM secret name on the Azure Key-Vault for the private SSH key.

code/cnf_operations/cnfoperations-upgrade.json@LIBCODE

- multiNodeOPM.opmMasterVMPrivateKeySecretName – MNOPM secret name on the Azure Key-Vault for the private SSH key.
- masterVmlpCopyCfgmgrPath – File path in master VM to place the backup of CNF cfgmgr. This should be a home folder of non-root user or any folder which has writing permission for non-root user. (i.e., /home/<non-root_username>)

code/cnf_upgrade_issu/cnfUpgradelssu.json@LIBCODE

- multiNodeOPM.opmMasterVMPrivateKeySecretName – MNOPM secret name on the Azure Key-Vault for the private SSH key.

Helm Delete Pipeline (Multinode OPM)

This pipeline deletes the multinode OPM using helm commands directed at the specified cluster.

pipeline/gmacro/helm-deletemacro-pipeline.yml@SVC-DEV

The pipeline contains the following variables:

- pool.name – Name of the Agent Pool.
- resources.repositories.repository – Name of repository used across the pipeline.
- resources.repositories.type – Type of repository.
- resources.repositories.name – Name of the actual ADO repository.
- resources.repositories.ref – Branch of the ADO repository.
- resources.containers.container – Container reference name which will be used across the pipeline.
- resources.containers.image – Image name with reference to ACR.
- zoneName – Name of zone. Used to name the pipeline run.
- template – Path to macrovariables.yml file in INFRA-DEV
- template – Path to nf-variables-templates.yml file in INFRA-DEV
- inputFile – Input JSON file name
- inputSourceFolder – Folder in LIB-CODE where inputFile is. Remove / at the end.

- `deployTimeout` – Time for the pipeline to run.
- `perFedMaxPollTimeout` – Max time for polling the fed which has been deployed.
- `finalValidationDeployTimeout` – Max time for validating the deployed fed.
- `finalValidationTimeout` - Maximum timeout in minutes until final polling occurs for all pods and services. After this, the Pipeline will exit with an error code if all pods and services are not running or complete.
- `stages.template` – Path to micro pipeline with corresponding ADO repository.

code/helm_deploy/helm_delete.json@LIB-CODE

This file contains the following variables as input to the Helm Deployment pipeline:

- `kubeconfig.reference.keyVault.id` - Name of the Azure Key-Vault.
- `kubeconfig.reference.keyVault.secretName` - Name of the secret in the Azure Key-Vault with value as kubeconfig of the Kubernetes Cluster.
- `prdLocation` - The folder location where the PRD files to deploy multinode OPM are present.
- `acrRepo` - The acr domain name where the images and charts are uploaded. (a4opadsacr.azurecr.io).
- `secretName` - The secret name to be provided which is created to pull the images and charts from acr repository (ex: acr-secret).
- `chartPath` - Relative path location where the charts are uploaded.
- `helmOperation` - Operation to be performed (<delete>).

Nexus CNF Operator Delete Pipeline

The Nexus CNF Operator Delete Pipeline has been enhanced to support the deletion of UPF.

nexusCnfOperatorDelete.json(LIB-CODE)

The following variable has been added:

- `cnf` - The CNF to be deleted. Enter smf or upf.

ACR to OPM Copy Pipeline

This pipeline now performs a pre-check to ensure images and charts have not been uploaded previously. Once this condition has been met, the ACRTTOOPM copy is triggered. If images and/or charts already exist in OPM, the pipeline does not run.

The pipeline also validates that images and charts have been successfully downloaded to OPM.

acrToOpmCopy.json

Contains the following new fields:

- opmDockerPorts - Multi-node OPM docker port to copy the images from ACR.
- opmDockerCharts- Multi-node OPM chart path where the charts are copied from the ACR.

For example:

```
"opmDockerPorts": [  
  "5006",  
  "5007",  
  "5008"  
],  
"opmDockerCharts": [  
  "chartm0",  
  "chartm1",  
  "chartm2"  
],
```

The pipeline uses these values to create a symlink to the opm.unitycloud.com:5005 folder by issuing the following command to copy the images and charts into the created folder, as shown here:

```
ln -s /etc/containerd/certs.d/<FQDN>:5005/  
/etc/containerd/certs.d/<FQDN>:5006  
ln -s /etc/containerd/certs.d/<FQDN>:5005/  
/etc/containerd/certs.d/<FQDN>:5007  
ln -s /etc/containerd/certs.d/<FQDN>:5005/
```

```
/etc/containerd/certs.d/<FQDN>:5008
```

New Features and Enhancements For 5.1.0.9

PADS VMISSU Download the tar file from storage account

As part of the feature a new stage has been introduced in the VM ISSU pipeline whose task is to download the tar file from the storage account and copied to the root directory of all the master and worker nodes.

New properties introduced as part of this features (**affirmed-nr-inServiceUpgrade.json**)

storageAccountResourceGroup : Resource group name where the tar file is uploaded

storageAccountName : Storage account name where the tar file is uploaded

containerName : Container name of the storage account

blobName : File name of the tar file.

NOTE : The assumption is that the tar file already uploaded to the storage account before attempting to trigger the VM ISSU Operation.

PADS Cinder CSI Support for CNF Deploy and Helm Deploy of MNOPM

A new file has been added "**affirmed-nr-cnf-operations-deploy-micro-pipeline-cinder-csi.yml**" under the location **/LIB-CODE/pipeline/cnfoperation** which will support the cinder csi storage creation.

A new file has been added "**affirmed-nr-helm-deploy-micro-pipeline-cinder-csi.yml**" under the location **/LIB-CODE/pipeline/helm_operation** which will support the cinder csi storage creation.

This stage will be run before the deployment of the CNF as well as helm deploy of MNOPM. Which will create the pure storage class.

New Features and Enhancements

The following properties below has been added to support the above functionality under the respective file name.

File Name : affirmed-nr-cnf-operations-deploy.json (**/LIB-CODE/code/cnfoperation**)

mechidPasswordSecretName : secret name present in the akv which holds the password of the mechid.

File Name : affirmed-nr-helm_deploy-multi-opm.json (**/LIB-CODE/code/helm_operation**)

mechidPasswordSecretName : secret name present in the akv which holds the password of the mechid.

The yaml file to support the pure storage creation is available at the location (**/LIB-CODE/code/cnfoperation**)

NOTE : Please change to the above micro pipeline file names while running for cnf operation and helm operation

Read k8 certificate key from the configparser override logs

As part of the fix the during the VM ISSU upgrade the k8 cert key is extracted from **the configParser_override.log** which was earlier read from the **configParser-<timestamp>.log**.

The following script file has been modified which will be included as part of the pads docker file.

- checkCerticate.sh
- upgradeMaster1.sh
- upgradeMaster2.sh

Finalizek8 cluster to support for the Old VM Image (5.1.0.6) for both CNF Operation and MNOPM operation.

As part of the feature, we have provided the backward compatibility of support the old VM Image 5.1.0.6.

The below files has been modified which are available under the respective path.

etcHostsCtrl_Old.sh (/LIB-CODE/code/finalize_cluster)

affirmed-nr-finalizeK8sCluster-micro-pipeline-old.yml (/LIB-CODE/pipeline/finalize_cluster)

affirmed-nr-mnopm-finalizeK8sCluster-micro-pipeline-old.yml (/LIB-CODE/pipeline/mnopm_finalize_cluster)

NOTE : Please change to the above micro pipeline file names while using the old VM Image to run the finalizek8cluster

New Features and Enhancements For 5.1.0.8

PADS one touch pipeline changes and enhancements for Nexus

The following is a summary of changes made as part of the AON PADS One Touch Pipeline Enhancements feature. Refer to the “Nexus Pipelines” section for configuration in the Affirmed Networks Packaging and Deployment Solutions Guide, Release 5.1.0-8 version.

Reorganization of Pipelines

Prior to version 5.1.0-8, the following macro pipelines were available across the SVC-DEV and INFRA-DEV repositories:

1. Publisher and Artifact Store Pipeline
2. Artifact Manifest Pipeline
3. ACR Copy artifacts Pipeline
4. Designer Pipeline
5. Operator CNF Deploy Pipeline
6. Operator CNF Delete Pipeline
7. Operator CNF ISSU Pipeline

These existing AON pipelines have been reorganized into the following pipelines, and the respective macro pipelines are now placed within the SVC-DEV repository.

1. Publisher Flow Pipeline
2. Designer Flow Pipeline

3. Operator CNF Deploy Pipeline
4. Operator CNF Delete Pipeline
5. Operator CNF ISSU Pipeline

All these pipelines are structured into different stages for each resource type.

Publisher Flow Pipeline

This pipeline is designed to create various AOSM resources, including:

- Publisher
- Artifact Store
- Network Function Definition Group (NFDG)
- Network Service Design Group (NSDG)

The macro pipeline file for the Publisher Flow Pipeline is located at:

SVC-DEV/pipeline/nexus-publisher/nexus-publisher-flow-macro-pipeline.yml

Designer Flow Pipeline

The Designer Flow Pipeline facilitates the creation of Artifact Manifests within the Publisher Artifact Store and the uploading of artifacts into the Artifact Store's Azure Container Registry (ACR). Furthermore, it creates AOSM resources such as:

- Network Function Definition Version (NFDV)
- Network Service Design Version (NSDV)
- CNF Specific Global Configuration Group Schema (Global CGS)
- Site-Specific Instance Configuration Group Schema (Instance CGS)
- Secret Configuration Group Schema (Secret CGS)

This enhancement merges the existing pipelines like Artifact Manifest, ACR Copy artifacts, and Designer into a single, one-touch Designer Flow Pipeline.

The macro pipeline file for the Designer Flow Pipeline is located at:

SVC-DEV/pipeline/nexus-designer/nexus-designer-flow-macro-pipeline.yml

Operator CNF Deploy Pipeline

This pipeline creates the following AOSM resources and deploys the CNF via creating SNS:

- CNF specific Global Configuration Group Value (Global CGV)

New Features and Enhancements

- Site-specific Instance Configuration Group Value (Instance CGV)
- Secret Configuration Group Value (Secret CGV)
- Site and deploys the CNF via creating Site Network Service (SNS)

The macro pipeline file for the Operator CNF Deploy Pipeline is located at:

SVC-DEV/pipeline/nexus-cnf_operator/nexus-cnf-operator-macro-pipeline.yml

Operator CNF Delete Pipeline

This pipeline deletes the following AOSM resources and deletes the CNF deployment by deleting the SNS:

- CNF specific Global Configuration Group Value (Global CGV)
- Site-specific Instance Configuration Group Value (Instance CGV)
- Secret Configuration Group Value (Secret CGV)
- Site
- Site Network Service (SNS)

The macro pipeline file for the Operator CNF Delete Pipeline is located at:

SVC-DEV/pipeline/nexus-cnf_operator/nexus-cnf-operator-delete-macro-pipeline.yml

Operator ISSU Pipeline

This pipeline creates or reuses the following AOSM resources and upgrades the CNF in existing site and existing Site Network Service (SNS):

- CNF specific Global Configuration Group Schema (Global CGS)
- Site-specific Instance Configuration Group Schema (Instance CGS)
- Secret Configuration Group Schema (Secret CGS)
- CNF specific Global Configuration Group Value (Global CGV)
- Site-specific Instance Configuration Group Value (Instance CGV)
- Secret Configuration Group Value (Secret CGV)

The macro pipeline file for the Operator CNF Delete Pipeline is located at:

SVC-DEV/pipeline/nexus-cnf_operator/nexus-cnf-operator-issu-macro-pipeline.yml

Separate bicep and parameters file for each AOSM resource

The existing bicep and parameters files in each pipeline are separated into individual bicep and parameters files for each AOSM resource in each pipeline.

Single place Pipeline Configuration

Two YAML files have been introduced in the SVC-DEV repository. JSON configurations in LIB-CODE for all pipelines now reference these files to update values before running. This approach consolidates the configuration for all pipelines in a single place, minimizing the need to update JSON configuration files in LIB-CODE.

1. Pipeline template variables file: This file consolidates all the pipeline parameters used across different pipelines.

File location: config/variables/pipeline-template-variables.yml@SVC-DEV

2. Azure Resource Name variables file: This file primarily configures Azure resource names for the publisher and designer flow pipelines, avoiding reliance on site-specific files since these pipelines can be executed ahead of time.

File location: config/variables/smf-ds-az-resourcename-template-variables.yml@SVC-DEV

Tags for AOSM Resources

Defining tags for AOSM resources is supported in the bicep file of each AOSM resource, driven by configuration data.

- Publisher and Designer Resources: Data is sourced from pipeline template variables and Azure resource name template variables files. The default tags include:
 - Correlation ID
 - Environment Type – Production or Non-production
 - Name of the Automator
 - Contact distribution list
 - Name of the application
 - Name of the program
 - Name of the vendor
 - UC Major Release Number (Use dash instead of dot; e.g., 4-3)
 - Build ID
 - Build Requested Email

New Features and Enhancements

- Build Definition Name
 - Program release name
- Operator Resources: Data is sourced from site-specific macro variables and the result JSON file. The default tags include:
 - Correlation ID.
 - Environment Type – Production or Non-production.
 - Name of the Automator.
 - Contact distribution list.
 - Type of Network Function.
 - AON Instance name.
 - NF Instance name.
 - Name of the vendor.
 - Software version.
 - Name of the Publisher

Pre and Post Deployment Checks for Operator CNF Deploy and ISSU

Added pre-deployment check to confirm custom location availability before deploying operator resources during Operator Deploy and CNF Operator ISSU pipelines.

Added post-deployment check to verify cluster node and pod status after deploying operator resources during Operator Deploy and CNF Operator ISSU pipelines.

These checks require the `kubect!` command. Make sure to install the `kubect!` command and add it to the PATH variable in the agent VM before running the pipelines.

Uncomment Code for IT Connection

Few lines of code to call connectit-send-notification-template.yaml and connectit-az-resource-status-template.yaml are commented out. ATT will uncomment these changes to connect IT before running the pipelines.

PRD Files Location

To hold all PRD files in a single location, a folder LIB-CODE/prd is created and configured in the JSON configuration files of different pipelines.

Segregation of Nexus and Network Cloud in Output Bundle

To separate Nexus and Network Cloud in the output bundle file pads_pipeline-<version>-<build_number>.tgz, two folders are created under pipeline-artifacts.tar. One is for Network Cloud and the other for Nexus. The respective pipeline code is found in these folders. Follow these instructions to set up the pipeline code:

Extract the pads_pipeline-<version>-<build_number>.tgz file. Extract the pipeline_artifacts.tar.gz file inside pads_pipeline-<version>-<build_number>/.

The results should show two directories: networkcloud/ and nexus/, along with third-party BOM and PRD files.

Inside the networkcloud and nexus directories, there should be three directories: LIB-CODE/, INFRA-DEV/, and SVC-DEV/.

Based on whether Network Cloud or Nexus is to be used, copy the contents of these directories under networkcloud or nexus extracted from pipeline_artifacts.tar.gz into the respective ADO repository. Refer to the “Install PADS Software” section for updated instructions in the Affirmed Networks Packaging and Deployment Solutions Guide, Release 5.1.0-8 version.

Naming convention for Operator AOSM resources

In Operator Deploy pipeline, name of AOSM resources to be created are configurable in Azure resource name template variables file. By default, these names are configured using “\$(alias)”. For example, “\$(alias)-cnf-sns-01” is configured for “snsName”. Here, \$(alias) will be replaced with value of “paramAlias” set in site-specific result json file before using in pipelines. These names are used in Operator ISSU pipeline when reusing existing resources.

VM ISSU Config Parameters and default value updated in bold (section added as per AT&T request)

Table 1. affirmed-nr-inServiceUpgrade.json (LIB-CODE) Variable

Variables	Description
opmMode	The type of OPM (single node/multi-node).
clusterName	The name of the Kubernetes cluster used for kubeconfig.
ucUpgraderConfig.vmType	The VM you are upgrading. (vm-opm or vm- platform) Default Value: vm- platform
ucUpgraderConfig.vmVersion	The VM version to which you are upgrading. Default Value: "vm-platform-5.1.0-13-rel-5-1-0.qcow2"

ucUpgraderConfig.seedFile	The seed File used to upgrade the VM. Default Value: "seed-vm-platform-5.1.0-19-rel-5-1-0.tgz"
ucUpgraderConfig.upgradeVM	Indicates if the VM is a new install (False) or is being upgraded (True). Default Value: True
ignoreConfigparserOverride	Set this value to True when the VM reboots after upgrade to cause the configParser to run as normal and ensure the upgrade will not stop after the IP address is configured. Default Value: True
ucUpgraderConfig.ucUpgraderConfig.k8sVersion	Change The Kubernetes version to which you are upgrading. If you enter the same version as what is currently configured, the Kubernetes version does not upgrade. Instead, the OS and kernel upgrade provided that the new seed file has the matching Kubernetes version Default Value: 1.29.7
ucUpgraderConfig.multiBootDebug	Set this variable to True so the multi-boot menu stays up for 4 minutes, then go to default partition. Set this variable to False to have the multi-boot menu time out after 30 seconds. Default Value: False
ucUpgraderTARFileURL	The URL to download tar package containing seed file and upgrade scripts Default Value: "https://tk5af.a4opacketcore.microsoft.com/artifactory/rel_build_artifacts/vms/vm-platform/ucUpgrader-vm-platform-5.1.0-19-rel-5-1-0.tar",
delayInUpgrade	Delay in minutes to upgrade each VM. Ideal time would be 5min. Value to be provided is 10m Default Value: 5m
forceErase	If True, script will not ask user if it is ok to delete old partition. Default Value: False
tarFileDestination	The location where the tar file want to read as input to the pipeline. Default Value: "/ucUpgrader-vm-platform-5.1.0-19-rel-5-1-0.tar"

New Features and Enhancements For 5.1.0.7

PADS one touch pipeline changes for Network Cloud

As part of the feature a new macro pipeline has been added in the SVC-DEV dev folder with file name "affirmed-nr-cnf-operations-fullbuild-macro-pipeline.yml"

pipeline/gmacro/affirmed-nr-cnf-operations-fullbuild-macro-pipeline.yml
@SVC-DEV

Input file which contains below variables as input to Full Build Pipeline.

Properties and parameters associated with the file are mentioned below

pool.name – Name of the Agent Pool.

resources.repositories.repository – Name of repository used across the pipeline.

resources.repositories.type – Type of repository.

resources.repositories.name – Name of the actual ADO repository.

resources.repositories.ref – Branch of the ADO repository.

resources.containers.container – Container reference name which will be used across the pipeline.

resources.containers.image – Image name with reference to ACR.

parameters :

zoneName – Name of zone. Used to name the pipeline run.

updateOpmSslCertificate - True if the OPM SSL certificate needs to be updated with an IP address. Note that for multi-node OPM, this variable has no-op. The multi-node OPM updates the OPM SSL Certs, and no action is required by PADS

createChartRepository - True to create a chart repository on the OPM used for CNF Operations

createDeploymentZone - True to create a Deployment Zone used to attach the K8s cluster from the FinalizeK8sCluster Pipeline.

updateRuntimeDeploytimeout - True to update runtime deploy timeout on the OPM.

variables :

New Features and Enhancements

template – Path to macrovariables.yml file in INFRA-DEV

template – Path to nf-variables-templates.yml file in INFRA-DEV

inputFile – Input JSON file name

inputSourceFolder – Folder in LIB-CODE where inputFile is. Remove / at the end.

deployTimeout – Time for the pipeline to run.

perFedMaxPollTimeout – Max time for polling the fed which has been deployed.

finalValidationDeployTimeout – Max time for validating the deployed fed.

finalValidationTimeout - Maximum timeout in minutes until final polling occurs for all pods and services. After this, the Pipeline will exit with an error code if all pods and services are not running or complete.

inputFileForOPMAutomation - Input json file for the OPM Automation pipeline

inputSourceFolderForOPMAutomation - Folder in LIB-CODE where inputFile is. Remove / at the end of the file path.

opmPrereqAutomationTimeout – Timeout of the OPM Automation pipeline

acrToOpmlImageCopyTimeout - Timeout in minutes for the pipeline acrtoopmcopy.

inputFileForACRToOPM - Input json file for ACRTOOPMCopy pipeline.

inputSourceFolderForACRToOPM - Folder in LIB-CODE where inputFile for inputFileForACRToOPM is. Remove / at the end of the file path.

finalizeK8ClusterTimeout - Timeout in minutes for the pipeline finalizek8cluster

operation – The value for the field is “finalizeK8sCluster”

inputSourceFolderForFinalizeCluster - Folder in LIB-CODE where inputFile for inputFileForFinalizeCluster

is. Remove / at the end of the file path.

inputFileForFinalizeCluster - Input json file for finalizek8Cluster pipeline.

isMultiNodeOPMClusterDeployment - The default value is “false” as we have a separate pipeline for MNOPM finalizek8Cluster

Affirmed CNF Deploy Operations Specific

fedDeployTimeout - Polling interval (in seconds) for calling NB API to poll all federations status. Recommended value is 10

cnfDeployTimeout - Total timeout in minutes for the CNF pipeline to run.

New Features and Enhancements

`finalValidationDeployTimeout` - Polling timeout in seconds for checking pods and services status for each fed

`finalValidationTimeout` - Maximum timeout in minutes until final polling occurs for all pods and services. After this, the Pipeline will exit with an error code if all pods and services are not running or complete

`siteSpecificSourceFolder` – Folder location where the site specific file required for cnf deploy operation present

`inputSourceFolderForCNFDeploy` - Folder in LIB-CODE where `inputFile` for `inputFileForCNFDeploy` is. Remove / at the end of the file path.

`inputFileForCNFDeploy` - Input json file for cnfdeploy pipeline.

`cnfOperation` – The default value is “deploy”.

`inputSourceFolderForFinalizeCluster` - Folder in LIB-CODE where `inputFile` for `inputFileForFinalizeCluster` is. Remove / at the end of the file path.

`inputFileForFinalizeCluster` - Input json file for finalizex8Cluster pipeline.

`macrovariablesSourceFolder` - Folder in INFRA-DEV where `inputFile` for macrovariable

`macrovariables` – Input json file for the macro variables.

`inputFileForCNFUpgrade` - Input json file for cnf upgrade pipeline.

`inputSourceFolderForCNFUpgrade` - Folder in LIB-CODE where `inputFile` for cnf upgrade pipeline is. Remove / at the end of the file path.

`stages.template` – Path to micro pipeline which are referencing in the LIB_CODE with corresponding ADO repository.

MNOPM Helm deploys changes to pull the certificates from the key vault

As part of these story the certificates of `fed-opm_multi.yml` and `fed-pass_helpers.yml` values file is uploaded in the base64 encoded format into the key vault .The secret name of the uploaded certificate is provided in the input json file. A bash script is provided in the pipeline which would replace the placeholders in the values.yaml file with the certificates from the key valut.

The following input values has been added in the `helm_deploy.json` file.

`code/helm_deploy/affirmed-nr-helm_deploy.json@LIB-CODE`

`OPMMultiTlsKeySecretName` : Secret name of tls key certificate for the `fed-opm_multi.yml` values file

`OPMMultiTlsCrtSecretName` : Secret name of tls certificate for the `fed-opm_multi.yml` values file

New Features and Enhancements

OPMMultiCaCrtSecretName: Secret name of ca certificate for the fed-opm_multi.yml values file

FedPaasHelpersTlsKeySecretName : Secret name of tls key certificate for the fed-pass_helpers.yml values file

FedPaasHelpersTlsCrtSecretName: Secret name of tls certificate for the fed-pass_helpers.yml values file

FedPaasHelpersCaCrtSecretName: Secret name of ca certificate for the fed-pass_helpers.yml values file

File Name Modification:

All the file name has been appended with the prefix “affirmed-nr” as shared by AT&T repos.

MNOPM FinalizeK8 Cluster Pipeline:

As part of these changes a new pipeline has been added to run the finalizek8cluster for MNOPM deployment.

The following sections show the configurable files in the MNOPM Finalize K8 Cluster Pipeline and their corresponding (repository):

- affirmed-nr-mnopm_finalizeK8sCluster-macro-pipeline.yml (INFRA-DEV)
- pipeline/mnopm_finalize_cluster/affirmed-nr-mnopm-finalizeK8sCluster-micro-pipeline.yml (LIB-CODE)
- code/mnopm_finalize_cluster/affirmed-nr-finalizeK8sCluster-multi-opm.json (LIB-CODE)

finalizeK8sCluster-macro-pipeline.yml (INFRA-DEV)

The following table shows the variables used in the FinalizeK8sCluster macro pipeline.

pool.name - Name of the Agent Pool

resources.repositories.repository- Name of the repository used across the pipeline

resources.repositories.type -Type of repository resources.repositories.name Name of the actual ADO repository

resources.repositories.ref - Branch of the ADO repository resources.containers.container Container reference name which will be used across the pipeline resources.containers.image -Image name with reference to ACR Pipelines variables

- azureSubscription - Azure subscription name

- inputFile - Input JSON file name
- inputSourceFolder - Folder in LIB-CODE where inputFile is. Remove / at the end of the file path
- template - ▪ The absolute path to the macrovariables.yml file in INFRADEV repository. When referring from another repository, add @INFRA at the end of the file path

stages.template - Path to micro pipeline with the corresponding ADO repository

name - Name of the Pipeline when it is run. Includes parameter zoneName

appendCommitMessageToRunName - Append the commit message to the build number. The default is true

zoneName - Parameter to specify zone name workerCount Parameter for number of workers

azureRegion - Parameter for the Azure region on which the Pipeline runs

code/mnopm_finalize_cluster/affirmed-nr-finalizeK8sCluster-multi-opm.json (LIB-CODE)

The following shows the variables used in the MNOPM FinalizeK8sCluster Input JSON pipeline.

opmMode - Specifies the OPM deployment type: singleNode or multiNode

multiNodeOPM.FQDN - For multi-node OPM deployment: FQDN of OPM configured when deploying multi-node OPM. Set the description value for this to opm.unitycloud.com.¹

Note: For additional information on how to deploy multi-node OPM with opm.unitycloud.com as the FQDN value and the Public certificate of multi-node OPM containing *.unitycloud.com, please refer to the UnityCloud documentation set.

multiNodeOPM.domainName - Domain name of the OPM FQDN multiNodeOPM.opmMasterVMIP - For multi-node OPM deployment: multimaster's VIP IP address (used to ssh to multinode OPM). Multi-master is where the multinode OPM is deployed.

multiNodeOPM.opmMasterVMPrivateKeySecretName - For multi-node OPM deployment: secret name in vault which contains the key that is used to ssh to master VM node without password.

attachCluster.clusterIP - NF cluster's master node VM IP address, used to attach a cluster to OPM.

¹

New Features and Enhancements

`attachCluster.m2mkeyKuberadminSecretName` - Name of the secret stored in the KeyVault secret name. PADS generates this name to store the NF cluster's M2M private key for the kuberadmin user. No input is required.

`attachCluster.attachType` - Specifies whether the cluster is openstack (`openstack_attach`) or baremetal (`baremetal_attach`). For non-Azure clusters, set this value to `nonazure_attach`.

`attachCluster.username` - Username OPM uses to SSH into the attaching K8 Cluster VMS to complete the attach cluster operation. You should use the root user name for this value.

`clusterName` - Name of the K8 cluster used for kubeconfig Pipelines Affirmed Networks Packaging and Deployment Solutions Guide Release 5.1.0 2-27 Revision: A Variables Description

`clusterParameters.ingressVM.adminAccess` - Communication type to Cluster. Accept the default value, SSH

`ClusterParameters.ingressVM.publicIpAddress`- IP-address to communicate with the jump VM

`ClusterParameters.ingressVM.port` - SSH port number to communicate with the jump VM

`ClusterParameters.ingressVM.adminUsername` - SSH username to communicate with the jump VM. Accept the default value, root

`ClusterParameters.ingressVM.adminPrivateKey.reference.keyVault.name` Azure Key-Vault name with SSH private-key to be used to get to the ingress-VM

`clusterParameters.ingressVM.AdminPrivateKey.reference.secretName` - Azure Key-Vault secret-name where the SSH private-key is used to get to the location of the VM

`clusterParameters.tuning` - True if Cluster tuning needs to be done. False otherwise

`ClusterParameters.vmM2MKey.reference.keyVault.name` Azure Key-Vault name with SSH private-key to be used from the ingress-VM to the individual VMS that forms the K8s cluster

`ClusterParameters.vmM2MKey`.

`reference.secretName` - Azure Key-Vault secret-name where the SSH private-key to be used from the ingress-VM to the individual VMs that form the K8s cluster is stored

`clusterParameters.vms[].ipAddress` - VM IP address

`clusterParameters.vms[].port` - VM SSH Port `clusterParameters.vms[].type` Type of Node (Master/Worker)

`clusterParameters.vms[].adminUsername` - SSH username to communicate with VMs

`clusterParameters.vms[].dataplane` - Set to true if the node (VM) must have a label `nodetype=dataplane`, otherwise set this value to false

`opm.ipAddress` - OPM IP address

`opm.sshUsername` - SSH username to communicate to OPM `opm.keyVault.name` Azure Key-Vault name with OPM information

opm.keyVault.opmPrivateKeysecretName - OPM secret name on the Azure Key-Vault for the private SSH key Pipelines

opm.certPath - Certificate path on OPM which copies to all VMs on K8s Cluster

certCopyPathToAllVMs - Path on each VM in cluster to where the certificate copies

opm.username - OPM GUI username

opm.opmPasswordSecretName - Azure Key-Vault secret name for OPM GUI user's password

opm.dzName - Deployment Zone name on OPM to which the create K8s Cluster needs to be attached

clusterParameters.ha_proxy_vip - The HA VIP address of the Kubernetes cluster post deployment

clusterParameters.k8_primary_master_ip IP address of the primary master node of the CNF Cluster

clusterParameters.vms[].racklabel - Following the CNF cluster creation, this value is used to label all nodes (rack and zone).

Commented Code in the pipeline which is to be run in ATT Site

The code below has been commented which is not run in the MSFT site. This need to be uncommented to make it run in the ATT Site

File : affirmed-nr-acr-to-opm-image-copy-micro-pipeline.yml

Under job:

```
- job: "PreChecks"
- bash: |
```

```
    # echo az account set --subscription $(JVS-AZ-NC-SUB) ATT
code is commented out as it is not required
```

```
    # az account set --subscription $(JVS-AZ-NC-SUB)
    # az account list -o table
    displayName: Setting account to ADO subscription
```

```
- job: "ACR_OPM"
```

```
  - bash: |
    # echo az account set --subscription $(JVS-AZ-NC-SUB)
    # az account set --subscription $(JVS-AZ-NC-SUB)
    # az account list -o table
    displayName: Setting account to ADO subscription
```

affirmed-nr-cnf-operations-deploy-micro-pipeline.yml

```
- job : 'cnfDeploy'
```

```
# Commented out the existing code as it is for ATT
```

```
  # - bash: |
  #       set -e;
```

New Features and Enhancements

```
#          echo "##[debug] Checking Keyvault Secrets..."
#
secretFilePath=$( _artifactsLocation)/input/${(json.secretValuesFile)};
#          echo "secretFilePath = $secretFilePath"
#          echo -e "\n"

#          echo "##[debug] iterate over the secret names(keys
in secretValues json object) in secret values json file"
#          for secretName in `jq -r '. | keys[] as $k |
"\($k)"' $secretFilePath`; do
#              echo "secretName = $secretName"
#              keyvaultSecretName=`jq --raw-output --arg tmp
"$secretName" '.[${tmp}]' $secretFilePath`;
#              output=$(az keyvault secret show --name
"$keyvaultSecretName" --vault-name
"${(json.kubeconfig.reference.keyVault.id)}" -o tsv)
#              if [[ -z $output ]]; then
#                  echo "##[error]Keyvault secret
$keyvaultSecretName not found. Exiting..."
#                  exit 1
#              else
#                  echo "##[section]$keyvaultSecretName secret
found."
#              fi
#              echo -e "\n"
#          done
#          displayName: "Check if secrets are present in Azure Key-
Vault"

# - bash: |
#       set -e;
#       echo "##[debug] Updating secret values file
$( _artifactsLocation)/input/${(json.secretValuesFile)}"
#
secretFilePath=$( _artifactsLocation)/input/${(json.secretValuesFile)};
#       echo "secretFilePath = $secretFilePath"
#       echo -e "\n"

#       # iterate over the secret names(keys in secretValues
json object) in secret values json file
#       for secretName in `jq -r '. | keys[] as $k |
"\($k)"' $secretFilePath`; do
#           echo "secretName = $secretName"
#           keyvaultSecretName=`jq --raw-output --arg tmp
"$secretName" '.[${tmp}]' $secretFilePath`;
#           echo -e "\n"

#           echo "##[debug] Fetching secret value from Azure
Key-Vault..."
#           keyvaultSecretValue=`az keyvault secret show --
name "$keyvaultSecretName" --vault-name
"${(json.kubeconfig.reference.keyVault.id)}" | jq -r '.value`;
#           echo "##[section] Successfully retrived value for
$keyvaultSecretName"
#           echo -e "\n"
```

New Features and Enhancements

```
#          echo "##[debug] Updating secret value in secret
values file..."
#          output=`jq --arg secretName "$secretName" --arg
keyvaultSecretValue "$keyvaultSecretValue"
"[\\"$secretName\\"]=\\"$keyvaultSecretValue\\" \"$secretFilePath`;
#          echo $output > $secretFilePath;
#          echo -e "\n"
#          done
#          displayName: "Replace secret names with values from
Azure Key-Vault"

# - bash: |
#       set -e;
#       echo "##[debug] Checking Snapshot file..."
#       export
secretSnapshotValuesFilePath="$(_artifactsLocation)/input/$(json.secretVal
uesFile)"
#       echo "secretSnapshotValuesFilePath =
$secretSnapshotValuesFilePath"
#       echo -e "\n"

#       echo "##[debug] Get snapshot variables from
$secretSnapshotValuesFilePath"
#       export snapshot_variables=( $(echo `jq -r '. |= keys
' $secretSnapshotValuesFilePath` | jq --raw-output '.[ ]') );
#       for snapshot_variable in "${snapshot_variables[@]}";
do
#           echo " variables: $snapshot_variable"
#           done
#           echo -e "\n"

#           echo "##[debug] Get snapshot variables values from
$secretSnapshotValuesFilePath"
#           for snapshot_variable in "${snapshot_variables[@]}";
do
#               export snapshot_variable_value=$(jq --raw-output
--arg tmp "$snapshot_variable" '.[ $tmp]' $secretSnapshotValuesFilePath |
sed 's/\\/\\\\\\\\\\\\\\\\/g');
#               sed -i
"s#$snapshot_variable#$snapshot_variable_value#g"
$(_artifactsLocation)/input/$(json.snapshotFile);
#               done
#               echo -e "\n"

#           echo "##[debug] Print updated snapshot file"
#           cat $(_artifactsLocation)/input/$(json.snapshotFile)
#           displayName: "Update Snapshot file with secret values"
#           timeoutInMinutes: 5
```

New Pipeline Added:

We have added the new pipeline **opm_readiness** , **opm_readiness_mnopm** **cleanup_cluster** which we found in the repo shared by ATT.

Note: These pipelines are not added by msft so we have not tested these pipelines at our end but have been included as part of the PADS repository deliverables.

Configuration changes

Upgrade Notes for PADS5.2.0

Helm Value Recommendations for MN-OPM PaaS Components

The fed-prometheus and fed-redis_cluster resource default settings can be set lower for use with fed-opm_multi. Below are the recommended settings:

fed-prometheus:

```
thanosQuery:
  resources:
    limits:
      cpu: "500m"
      memory: 1Gi
    requests:
      cpu: "100m"
      memory: 256Mi
```

```
prometheus:
  resources:
    limits:
      cpu: "500m"
      memory: 1Gi
    requests:
      cpu: "100m"
      memory: 256Mi
```

fed-redis_cluster:

```
resources:
  limits:
    cpu: 500m
    memory: 1Gi
  requests:
    cpu: 100m
    memory: 512Mi
```

PADS Features

ADO-1296824 - PADS Support of NF Level Rollback-on-Failure

Release Note/User guide name	Section in Release Note /user guide	Any other comments
PADS_ReleaseNotes_5.2.0.docx	PADS Support of NF Level Rollback on CNF ISSU failure in Nexus	
PADS_UserGuide_Release_5.2.0.docx	Configure pipeline template variables, Sample CNF ARM template, Nexus CNF Operator ISSU Pipeline	

ADO-1656357 - [AT&T] Unity Cloud - Support for Cinder CSI, Encryption and rotation required for cloud provider passwords in heat template

Release Note/User guide name	Section in Release Note /user guide	Any other comments
PADS_ReleaseNotes_5.2.0.docx	PADS Cinder CSI Support for CNF Deploy and Helm Deploy of MNOPM	
PADS_UserGuide_Release_5.2.0.docx	Changes to be made in the SVC-DEV file to run Cinder CSI pipeline	

ADO-1558451 - [PADS] Integrate HashiCorp Vault for handling secrets used in PADS workflow flow for both NC and Nexus deployment

Release Note/User guide name	Section in Release Note /user guide	Any other comments
PADS_ReleaseNotes_5.2.0.docx	HashiCorp Vault Integration with MNOPM	
PADS_UserGuide_Release_5.2.0.docx	HashiCorp Vault Integration with MNOPM, Helm Deployment Pipeline, Helm Update Pipeline	

Known Environment Issues and Mitigation

Here are a few typical issues that could be faced due to environment / setup issues like network connectivity, permission etc. The section captures such issues and mitigation actions.

Typical Environment Issues

Anomaly	Description
Network connectivity issues cause errors when running the ACR to OPM Image Copy pipeline	The ACR to OPM Image Copy pipeline gives a <code>client_loop: send disconnect: Broken pipe</code> error when it is run. Workaround: <i>Re-run the pipeline.</i>
Network Connectivity issue causes CNF Upgrade pipeline to fail in the CNF Delete stage	The CNF Upgrade pipeline may fail during the CNF Delete stage. The following error may also appear: <pre>Other error occurred: HTTPSConnectionPool (host='opm.com', port=443): Max retries exceeded with url: /opm/nb/allFederations/mnopm-pads-dz/** (Caused by ConnectTimeoutError (<urllib3.connection.HTTPSConnection object at 0x7fafbf867ac0>, 'Connection to opm.com timed out. (connect timeout=None)'))</pre> Workaround: <i>Re-run the failed stage(s) of the pipeline.</i>
Permissions-related issues occur when pushing images to the AOSM Artifacts Store's ACR using the <code>cna_deploy_inventory</code> script when running the Artifact Manifest Pipeline.	When pushing images to the AOSM Artifact Store's ACR using the <code>cna_deploy_inventory</code> script, a permissions issue related to <code>az acr</code> commands may occur Workaround: <i>Add the related Service Principal as a contributor to the AOSM Artifact Store's ACR. From the Azure portal, navigate to the AOSM Artifact Store's ACR. Select Access Control then + Add Role Assignment.</i>
OPM NB API for attaching clusters must support non-root users	The <code>finalizeK8sCluster</code> pipeline invokes OPM NB API for attaching Cluster to OPM. As this API does not support non-root users, attaching cluster to OPM fails with "Unable to connect the cluster" error. Workaround: <i>Configure the NF master-1 node IP as "attachCluster.clusterIP" in <code>code/finalize_cluster/finalizeK8sCluster.json@LIB-CODE</code> instead of NF master VIP before running the <code>finalizeK8sCluster</code> pipeline.</i>

Known Anomalies

This section lists the known anomalies in this release.

Note

- Ensure that any Service Principal used for executing Nexus Pipelines to create AOSM artifacts in a given Resource Group is granted Contributor level access to that Resource Group.
- If a Nexus-related PRD must be modified, complete this before running them with Nexus pipelines. Managing PRD files is beyond the scope of the PADS software application.

Current Open Issues

This release contains the following known anomalies.

Table 2. Current Open Issues

Anomaly	Description
Only upgrading to higher version is supported in VM ISSU (1656345)	VM ISSU support upgrading to higher VM Images like 5.1.0.7 to 5.1.0.8
Pods stuck at image pullback when using AOSM local registry (2009636)	After SNS put/Reput the NF workload is not coming up as the pods fail to pull images from local registry when using cluster registry in Nexus clusters
CNF ISSU failure wait for timeout (1664358)	If any failure happens during the CNF ISSU .User need to wait for the timeout to happen from the OPM to trigger the next pipeline
CNF ISSU health checks for ceph , redis before starting ISSU via PADS (1827166)	This is an improvement request from internal testing to add pre-checks for ceph ,redis

ATT Open Enhancement Requests

The following enhancement requests are under assessment / planned as of now

Table 3. Open Enhancements

Anomaly	Notes / Planned Release
All links in current Pipelines need to refer to AT&T ACR not MSFT JFROG/SFTP	Under Discussion to use Storage account – planned for 5.1.0.9
Cinder CSI MOP automation via PADS	Collaborative development planned to deliver by 5.1.0.9 release
Improved logging and validations for the 4.3 production issues	Planned for 5.1.0.9
VM ISSU for OPM and OPM ISSU	Feature request from ATT pending.

PADS/OPM/UC Supportability Matrix

The following table shows the overview of the supportability between various versions of PADS, OPM, OPM VM, and UC:

PADS Version	OPM Version	OPM VM Version	UC Version
PADS 1.4.2	OPM 3.2		UC3.2, UC 3.3
PADS 1.4.3	OPM 3.3 "fed-opm-3.4.0-19-rel-3-3-0"	"vm-opm-3.2.0-19-rel-3-1-1.qcow2"	U 3.2, UC 3.3
PADS 4.1	OPM 4.1 "fed-opm-4.1.0-10-rel-4-1-0.tgz"	"vm-opm-3.4.0-206-main-int.qcow2"	UC 3.3, UC 4.1
PADS 4.2	OPM 4.2 - single Node OPM fed-opm-4.2.0-48-rel-4-2-0	"vm-opm-3.4.0-206-main-int.qcow2"	IJC 3.3, UC 4.1, UC 4.2
PADS 4.2	OPM 4.2.0.3 - single Node OPM fed-opm-4.2.0-51-rel-4-2-0	"vm-opm-3.4.0-206-main-int.qcow2"	UC 4.2.0.3
PADS 4.2.0.4	OPM 4.2.0.3 - single Node OPM	"vm-opm-3.4.0-206-main-int.qcow2"	UC 3.3, UC 4.1, UC 4.2
PADS 4.2.0.4	OPM 4.2.0.3 - Multi Node OPM fed-opm_multi-4.2.0-59-rel-4-2-0	"vm-opm-3.4.0-206-main-int.qcow2"	UC 4.1, UC 4.2 UC 3.3 not supported (container D issue)
PADS 4.3.0	opm-4.3.0-42-rel-4-3-0	"vm-opm-4.3.0-222-main-int-volsnapshot"	UC 4.3.0
PADS 4.3.0.3	opm-4.3.0-60-rel-4-3-0	"vm-opm-4.3.0-222-main-int-volsnapshot"	UC 4.3.0-202
PADS 5.1.0	OPM 5.1.0.x	"vm-opm-0.0.0-242-main-int-volsnapshot", "vm-opm-0.0.0-5-rel-5-1-0-volsnapshot", "vm-opm-5.1.0-12-rel-5-1-0-volsnapshot", "vm-opm-5.1.0-15-rel-5-1-0-volsnapshot", "vm-opm-5.1.0-16-rel-5-1-0-volsnapshot"	UC 4.2, UC 5.1.0.x

Problems Fixed

This release contains the following corrections.

Fixes in 5.1.0.8

The following items were addressed in this release.

Table 4. Problems Fixed in 5.1.0.8

Anomaly	Description
[AT&T]: SHA value missing/invalid for helm-charts in BOM	Updated the bom file to fix the issue with the mapping PRD to support uploading of the charts and images to ACR
MN OPM deployment failure with 5108 VM Images changes	The path of the user data has been changed in the new VM Image so updated them in finalizek8 cluster of MNOPM and CNF deploy
E2E-SOAK:PADS-VM ISSU pipeline should upload ucUpgrader tar to all VMs	Added the new functionality to pull the ucUpgrader tar file from the url mentioned in the input json before VM upgrade
VM ISSU Dev : On re-run of pipeline Post VM upgrade success and pipeline failure, pipeline attempts and fails same VM upgrade again	If the pipeline is rerun if the node is already upgraded to new version, it will ignore and proceed to next VM
ATT Production issue supported for PADS (5.1.0)	All the production issue observed in the 4.2 at ATT site has been addressed (Cluster lock issue, Pulling the certificates from akv)
ATT issue with finalizek8cluster worker nodes going to not ready state(5.1.0.8)	Additional validation check has been done to add the configuration to kubernetes config.yaml. Configuration is added only if it is not present
Vulnerability: GO (Go) Security Update for github.com/docker/docker (GHSA-v23v-6jw2-98fq) - Vulnerability Id and Image: 5000507_cna-pads	Helm and kubernetes client version has been updated in the PADS to address the vulnerability

Fixes in 5.1.0.7

The following items were addressed in this release.

Table 5. Problems Fixed in 5.1.0.7

Anomaly	Description
A 500 error occurs when polling the deploy and delete pipelines.	If we get any error from OPM we are exiting the pipeline with the error message.
Pads to support helm deploy to replace the certificate secrets from keyvault.	Helm deploy pipeline to support replacement of certificates from akv to the values.yaml file .

Fixes in 5.1.0.5

The following items were addressed in the 5.1.0.5 release.

Table 6. Problems Fixed in 5.1.0.5

Anomaly	Description
ADO 1619976: MultiNode OPM - Validate and Downloading OPM Public Certificate fix for AT&T	Fixed the issue appearing during the validating the MNOPM Certificate in the pipeline (OPM Automation,CNF Deploy,FinalizeK8 cluster and Delete pipeline)
ADO 1625371: ATT: 5.1.0.3 PADS pipeline is skipping precheck and download images stages in acr-opm pipeline	Fixed the issue happening while running the ACRTOPMCPY pipeline for single node
ADO 1630694: ATT: Incorrect status for upgrade/rollback ISSU failures	Fixed the issue with failure/error indication and stop the pipeline if deploy and rollback fails with the CNF ISSU upgrade pipeline.
ADO 1619978: Remove nfType from finalizek8 cluster as it is part of global variables	Removed the nfType variable in the finalizek8cluster

Fixes in 5.1.0.3

The following items were addressed in the 5.1.0.3 release.

Table 7. Problems Fixed in 5.1.0.3

Feature	Description
Nexus ACR Copy Artifacts Pipeline	The pipeline/nexus_acr_copy_artifacts/nexus-acrcopy-artifacts-micropipeline.yml (LIBCODE) pipeline failed with "command not found" error. The error occurred when running ACR to ACR copy artifacts on the Nexus ACR Copy Artifact Pipeline. This error has been corrected by ensuring there is no space following the equal sign in the nexus-acr-copyartifacts-micro-pipeline.yml file: Task: Download and Push Helm Charts to ACR <pre> line 202: sourceAcrName=\${sourceAcrName//\"} line 203: sourceAcrName=\${sourceAcrName//\' } line 206: helm registry login \${sourceAcrName}.azurecr.io --username "\$sourceUsername" --password "\$sourcePassword"; </pre>

Install PADS Software

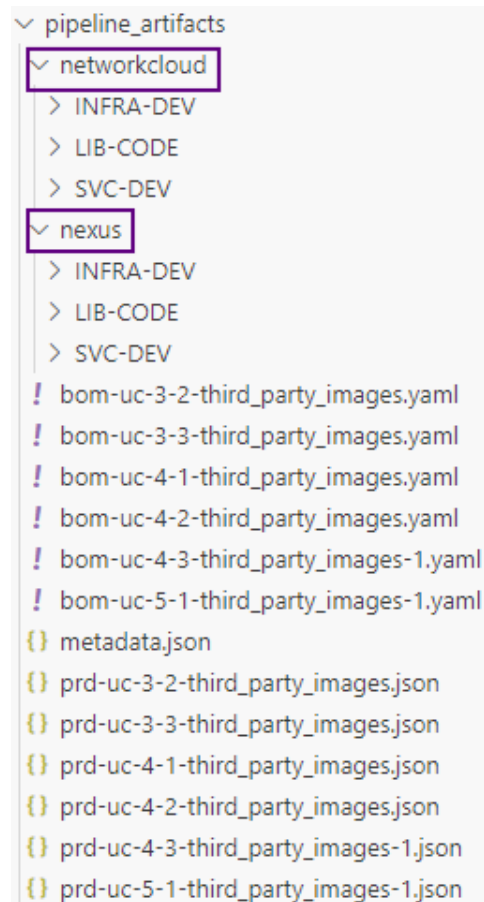
Prerequisites

Verify that the following prerequisite tasks are complete prior to installing the PADS software:

- Extract the `pads_pipeline-<version>-<build_number>.tgz` file. You will get the folder `pads_pipeline-<version>-<build_number>/`.
- Inside `pads_pipeline-<version>-<build_number>` folder, you will find the `pipeline_artifacts.tar.gz` file.
- Extract the `pipeline_artifacts.tar.gz` file.

You will get two directories `networkcloud/` and `nexus/` and BOM and PRD files.

Inside `networkcloud` and `nexus` directories, there should be 3 directories each - `LIB-CODE/`, `INFRA-DEV/` and `SVC-DEV/`.



Install PADS Software

- Confirm that the following repositories are created on the ADO project:
 - LIB-CODE
 - INFRA-DEV
 - SVC-DEV
- Verify that Service Connections to the following are created on the ADO project. (Reference to the [instructions for creating service connections](#))
 - a. Azure Resource Group
 - b. Azure Container Registry

Steps to Install

Network cloud

Complete the following steps in each ADO repository to install.

LIB-CODE

1. Copy the contents of the `networkcloud/LIB-CODE/` directory extracted from `pipeline_artifacts.tar.gz` into the LIB-CODE ADO repository.
2. Update the `code/acr_opm_image_copy/acrToOpmCopy.json` file with appropriate values.
3. Update the `code/finalize_cluster/finalizeK8sCluster.json` file with appropriate values.
4. Update the `code/opm_automation/opmAutomation.json` file with appropriate values.
5. Update the `code/nexus_cnf_operator/nexusCnfOperator.json` file with appropriate values.
6. Update the following files with the appropriate files.
 - `code/cnf_operations/cnf-operations-deploy.json`
 - `code/cnf_operations/cnf-operations-delete.json`
 - `code/cnf_operations/cnf-operations-upgrade.json`
 - `code/cnf_operations/cnf-operations-status.json`
7. Add the `cna_deploy_inventory` script file to a folder in LIB-CODE and refer to it in the appropriate variables.

Note: *UnityCloud version 4.1 of the `cna_deploy_inventory` script must be used.*

8. Add the *snapshotFile* value and *snapshot_specific_values.json* files to LIB-CODE and refer to them in the appropriate variables.

INFRA-DEV

1. Copy the contents of the INFRA-DEV/ directory extracted from pipeline_artifacts.tar.gz into the INFRA-DEV ADO repository.
2. Update the values in the following macro pipeline files: *acr-to-opm-image-copy-macropipeline.yml*, *finalizeK8sCluster-macro-pipeline.yml*, *opm-automation-macro-pipeline.yml*, *pipelineLocking-macro-pipeline.yml*, and *csoSoftwareInstall-macro-pipeline.yml*.

SVC-DEV

1. Copy the contents of the SVC-DEV/ directory extracted from pipeline_artifacts.tar.gz into the SVC-DEV ADO repository.
2. Update the values in the macro pipeline files: *cnf-operations-delete-macro-pipeline.yml*, *cnfoperations-deploy-macro-pipeline.yml*, *cnf-operations-upgrade-macro-pipeline.yml*, and *cnfoperations-status-macro-pipeline.yml*.

Nexus

Complete the following steps in each ADO repository to install.

LIB-CODE

1. Copy the contents of the nexus/LIB-CODE/ directory extracted from pipeline_artifacts.tar.gz into the LIB-CODE ADO repository.
2. The following configuration files are already updated with default values. Update them if required.
 - LIB-CODE/code/nexus_acr_copy_artifacts/affirmed-ds-smf-nexusAcrCopyArtifacts.json
 - LIB-CODE/code/nexus_acr_copy_artifacts/affirmed-ds-upf-nexusAcrCopyArtifacts.json
 - LIB-CODE/code/nexus_artifact_manifest/affirmed-ds-smf-nexusArtifactManifest.json
 - LIB-CODE/code/nexus_artifact_manifest/affirmed-ds-upf-nexusArtifactManifest.json
 - LIB-CODE/code/nexus_cnf_operator/affirmed-ds-smf-nexusCnfOperator.json

- LIB-CODE/code/nexus_cnf_operator/affirmed-ds-smf-nexusCnfOperatorDelete.json
- LIB-CODE/code/nexus_cnf_operator/affirmed-ds-upf-nexusCnfOperator.json
- LIB-CODE/code/nexus_cnf_operator/affirmed-ds-upf-nexusCnfOperatorDelete.json
- LIB-CODE/code/nexus_cnf_operator_issu/affirmed-ds-smf-nexusCnfOperatorIssu.json
- LIB-CODE/code/nexus_cnf_operator_issu/affirmed-ds-upf-nexusCnfOperatorIssu.json
- LIB-CODE/code/nexus_designer/affirmed-ds-smf-nexusDesigner.json
- LIB-CODE/code/nexus_designer/affirmed-ds-upf-nexusDesigner.json
- LIB-CODE/code/nexus_publisher/affirmed-ds-smf-nexusPublisher.json
- LIB-CODE/code/nexus_publisher/affirmed-ds-upf-nexusPublisher.json

3. Update the following files with appropriate values.

- code/nexus_designer/aosm/nfd/schemas/nfDeployparameters.json
- code/nexus_designer/aosm/nfd/values/values.json
- code/nexus_designer/aosm/nsd/schemas/secretCGSchema.json
- code/nexus_designer/aosm/nsd/schemas/siteSpecificCGSchema.json
- code/nexus_designer/aosm/nsd/schemas/smfCGSchema.json
- code/nexus_designer/aosm/nsd/schemas/upfCGSchema.json
- code/nexus_cnf_operator/aosm/cnf_nsd_values/cnfCGV.json
- code/nexus_cnf_operator/aosm/cnf_nsd_values/secretCGV.json
- code/nexus_cnf_operator/aosm/cnf_nsd_values/siteSpecificCGV.json
- code/nexus_cnf_operator_issu/aosm/cnf_nsd_values/cnfCGV.json
- code/nexus_cnf_operator_issu/aosm/cnf_nsd_values/secretCGV.json
- code/nexus_cnf_operator_issu/aosm/cnf_nsd_values/siteSpecificCGV.json
- code/nexus_cnf_operator_issu/aosm/nfd/schemas/nfDeployparameters.json
- code/nexus_cnf_operator_issu/aosm/nfd/values/values.json
- code/nexus_cnf_operator_issu/aosm/nsd/schemas/secretCGSchema.json
- code/nexus_cnf_operator_issu/aosm/nsd/schemas/siteSpecificCGSchema.json

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- code/nexus_cnf_operator_issu/aosm/nsd/schemas/smfCGSchema.json
 - code/nexus_cnf_operator_issu/aosm/nsd/schemas/upfCGSchema.json
4. Place the PRD files under LIB-CODE/prd folder and configure the PRD file names in appropriate pipeline-template-variables.yml file under SVC-DEV.
 5. Place the CNF ARM template file (smf_nf_template_artifact.json or upf_nf_template_artifact.json) under LIB-CODE/code/nexus_acr_copy_artifacts/armTemplateFolder
 6. Place the *cna_deploy_inventory* script under LIB-CODE/code/nexus_acr_copy_artifacts/scripts.

Note: UnityCloud version 4.1 of the *cna_deploy_inventory* script must be used. **TODO: Update the version**

INFRA-DEV

1. Note that nexus/INFRA-DEV folder extracted from pipeline_artifacts.tar.gz will be empty.
2. Create the following files under INFRA-DEV repository and configure their paths appropriately in operator macro pipeline files.
 - Site specific macro variables file
 - Site specific result json configuration file
 - CNF version specific json configuration file

SVC-DEV

1. Copy the contents of the nexus/SVC-DEV/ directory extracted from pipeline_artifacts.tar.gz into the SVC-DEV ADO repository.
2. Update the values in the following variables files:
 - SVC-DEV/config/variables/smf-ds-az-resourcename-template-variables.yml
 - SVC-DEV/config/variables/smf-ds-pipeline-template-variables.yml
 - SVC-DEV/config/variables/upf-ds-az-resourcename-template-variables.yml
 - SVC-DEV/config/variables/upf-ds-pipeline-template-variables.yml
3. Update the values in the following macro pipeline files:
 - SVC-DEV/pipeline/nexus-publisher/nexus-publisher-flow-macro-pipeline.yml
 - SVC-DEV/pipeline/nexus-designer/nexus-designer-flow-macro-pipeline.yml
 - SVC-DEV/pipeline/nexus-cnf_operator/nexus-cnf-operator-macro-pipeline.yml

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- SVC-DEV/pipeline/nexus-cnf_operator/nexus-cnf-operator-issu-macro-pipeline.yml
- SVC-DEV/pipeline/nexus-cnf_operator/nexus-cnf-operator-delete-macro-pipeline.yml

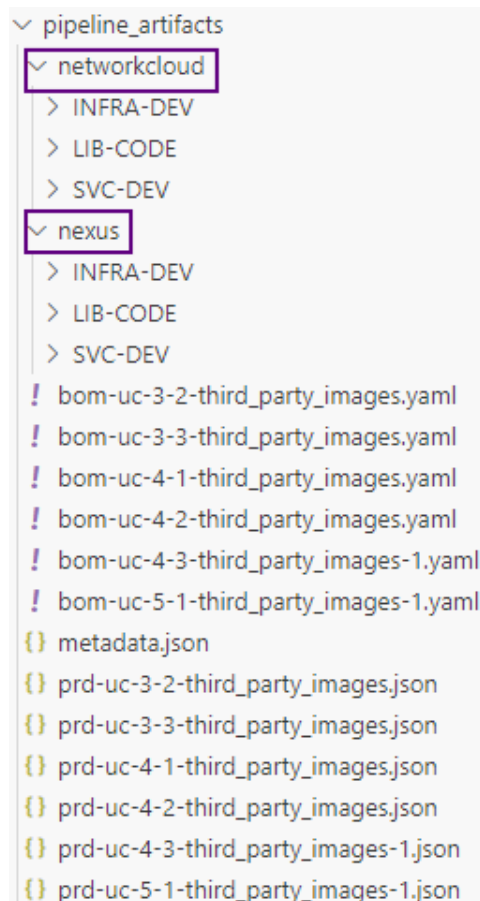
Upgrade PADS Software

Prerequisites

Verify that the following prerequisite tasks are complete prior to installing the PADS software:

- Extract the `pads_pipeline-<version>-<build_number>.tgz` file. You will get the folder `pads_pipeline-<version>-<build_number>/`.
- Inside `pads_pipeline-<version>-<build_number>` folder, you will find the `pipeline_artifacts.tar.gz` file.
- Extract the `pipeline_artifacts.tar.gz` file.

You will get two directories `networkcloud/` and `nexus/` and BOM and PRD files.



Inside `networkcloud` and `nexus` directories, there should be 3 directories each - `LIB-CODE/`, `INFRA-DEV/` and `SVC-DEV/`.

- Confirm that the following repositories are created on the ADO project:
 - LIB-CODE
 - INFRA-DEV
 - SVC-DEV
- Verify that Service Connections to the following are created on the ADO project. (Reference to the [instructions for creating service connections](#))
 - a. Azure Resource Group
 - b. Azure Container Registry

Steps to Upgrade

Network cloud

Complete the following steps in each ADO repository to upgrade.

LIB-CODE

1. Override all files in `pipeline/` with the contents of `networkcloud/LIB-CODE/pipeline/` directory extracted from `pipeline_artifacts.tar.gz`.
2. Add new variables to `code/acr_opm_image_copy/acrToOpmCopy.json`.
 - Use `LIB-CODE/code/acr_opm_image_copy/acrToOpmCopy.json` extracted from `pipeline_artifacts.tar.gz` for reference.
3. Add new variables to `code/finalize_cluster/finalizeK8sCluster.json`.
 - Use `LIB-CODE/code/finalize_cluster/finalizeK8sCluster.json` extracted from `pipeline_artifacts.tar.gz` for reference.
4. Add new variables to `code/opm_automation/opmAutomation.json`.
 - Use `LIB-CODE/code/opm_automation/opmAutomation.json` extracted from `pipeline_artifacts.tar.gz` for reference.
5. Add new variables to `code/nexus_cnf_operator/nexusCnfOperator.json`.
 - Use `LIB-CODE/code/nexus_cnf_operator/nexusCnfOperator.json` extracted from `pipeline_artifacts.tar.gz` for reference.

6. Update the following files with appropriate variables from the Affirmed Networks Packaging and Deployment Solutions Guide. Use the corresponding LIB-CODE/ directory files for reference.

- *code/cnf_operations/cnf-operations-deploy.json*
- *code/cnf_operations/cnf-operations-delete.json*
- *code/cnf_operations/cnf-operations-upgrade.json*
- *code/cnf_operations/cnf-operations-status.json*

7. Add the *cna_deploy_inventory* script file to a folder in LIB-CODE and refer to the script file.

Note: *UnityCloud version 4.1 of the cna_deploy_inventory script must be used.*

8. Add the PRD files to *code/acr_opm_image_copy* and refer to the PRD files.

INFRA-DEV

Update the values in the following macro pipeline files: *acr-to-opm-image-copy-macropipeline.yml*, *finalizeK8sCluster-macro-pipeline.yml*, *opm-automation-macro-pipeline.yml*, *pipelineLocking-macro-pipeline.yml*, and *csoSoftwareInstall-macro-pipeline.yml*.

SVC-DEV

Update the values in the macro pipeline files: *cnf-operations-delete-macro-pipeline.yml*, *cnfoperations-deploy-macro-pipeline.yml*, *cnf-operations-upgrade-macro-pipeline.yml*, and *cnfoperations-status-macro-pipeline.yml*.

Nexus

Complete the following steps in each ADO repository to upgrade.

LIB-CODE

1. Take a backup of LIB-CODE ADO repository and update the files as in the next steps based on this backup.
2. Overwrite the contents of the nexus/*LIB-CODE*/ directory extracted from pipeline_artifacts.tar.gz into the LIB-CODE ADO repository.
3. The following configuration files are already updated with default values. Update them if required.
 - LIB-CODE/code/nexus_acr_copy_artifacts/affirmed-ds-smf-nexusAcrCopyArtifacts.json
 - LIB-CODE/code/nexus_acr_copy_artifacts/affirmed-ds-upf-nexusAcrCopyArtifacts.json
 - LIB-CODE/code/nexus_artifact_manifest/affirmed-ds-smf-nexusArtifactManifest.json
 - LIB-CODE/code/nexus_artifact_manifest/affirmed-ds-upf-nexusArtifactManifest.json
 - LIB-CODE/code/nexus_cnf_operator/affirmed-ds-smf-nexusCnfOperator.json
 - LIB-CODE/code/nexus_cnf_operator/affirmed-ds-smf-nexusCnfOperatorDelete.json
 - LIB-CODE/code/nexus_cnf_operator/affirmed-ds-upf-nexusCnfOperator.json
 - LIB-CODE/code/nexus_cnf_operator/affirmed-ds-upf-nexusCnfOperatorDelete.json
 - LIB-CODE/code/nexus_cnf_operator_issu/affirmed-ds-smf-nexusCnfOperatorIssu.json
 - LIB-CODE/code/nexus_cnf_operator_issu/affirmed-ds-upf-nexusCnfOperatorIssu.json
 - LIB-CODE/code/nexus_designer/affirmed-ds-smf-nexusDesigner.json
 - LIB-CODE/code/nexus_designer/affirmed-ds-upf-nexusDesigner.json
 - LIB-CODE/code/nexus_publisher/affirmed-ds-smf-nexusPublisher.json
 - LIB-CODE/code/nexus_publisher/affirmed-ds-upf-nexusPublisher.json
4. Update the following files with appropriate values using backup
 - code/nexus_designer/aosm/nfd/schemas/nfDeployparameters.json
 - code/nexus_designer/aosm/nfd/values/values.json
 - code/nexus_designer/aosm/nsd/schemas/secretCGSchema.json
 - code/nexus_designer/aosm/nsd/schemas/siteSpecificCGSchema.json
 - code/nexus_designer/aosm/nsd/schemas/smfCGSchema.json
 - code/nexus_designer/aosm/nsd/schemas/upfCGSchema.json

- code/nexus_cnf_operator/aosm/cnf_nsd_values/cnfCGV.json
 - code/nexus_cnf_operator/aosm/cnf_nsd_values/secretCGV.json
 - code/nexus_cnf_operator/aosm/cnf_nsd_values/siteSpecificCGV.json
 - code/nexus_cnf_operator_issu/aosm/cnf_nsd_values/cnfCGV.json
 - code/nexus_cnf_operator_issu/aosm/cnf_nsd_values/secretCGV.json
 - code/nexus_cnf_operator_issu/aosm/cnf_nsd_values/siteSpecificCGV.json
 - code/nexus_cnf_operator_issu/aosm/nfd/schemas/nfDeployparameters.json
 - code/nexus_cnf_operator_issu/aosm/nfd/values/values.json
 - code/nexus_cnf_operator_issu/aosm/nsd/schemas/secretCGSchema.json
 - code/nexus_cnf_operator_issu/aosm/nsd/schemas/siteSpecificCGSchema.json
 - code/nexus_cnf_operator_issu/aosm/nsd/schemas/smfCGSchema.json
 - code/nexus_cnf_operator_issu/aosm/nsd/schemas/upfCGSchema.json
5. Place the PRD files under LIB-CODE/prd folder and configure them in appropriate pipeline-template-variables.yml file under SVC-DEV
 6. Place the CNF ARM template file (smf_nf_template_artifact.json or upf_nf_template_artifact.json) under LIB-CODE/code/nexus_acr_copy_artifacts/armTemplateFolder
 7. Place the *cna_deploy_inventory* script under LIB-CODE/code/nexus_acr_copy_artifacts/scripts

Note: *UnityCloud version 4.1 of the cna_deploy_inventory script must be used.*

INFRA-DEV

1. Note that nexus/INFRA-DEV folder extracted from pipeline_artifacts.tar.gz will be empty
2. Take a backup of INFRA-DEV ADO repository and update the files in the next steps based on this backup.
3. Make the INFRA-DEV ADO repo empty.
4. Create the following files under INFRA-DEV folder and configure their paths appropriately in operator macro files.
 - Site specific macro variables file
 - Site specific result json configuration file

- CNF version specific json configuration

SVC-DEV

1. Take a backup of INFRA-DEV ADO repository and update the files in the next steps based on this backup.
2. Overwrite the contents of the nexus/SVC-DEV/ directory extracted from pipeline_artifacts.tar.gz into the SVC-DEV ADO repository.
3. Update the values in the following variables files.
 - SVC-DEV/config/variables/smf-ds-az-resourcename-template-variables.yml
 - SVC-DEV/config/variables/smf-ds-pipeline-template-variables.yml
 - SVC-DEV/config/variables/upf-ds-az-resourcename-template-variables.yml
 - SVC-DEV/config/variables/upf-ds-pipeline-template-variables.yml
4. Update the values in the following macro pipeline files:
 - SVC-DEV/pipeline/nexus-publisher/nexus-publisher-flow-macro-pipeline.yml
 - SVC-DEV/pipeline/nexus-designer/nexus-designer-flow-macro-pipeline.yml
 - SVC-DEV/pipeline/nexus-cnf_operator/nexus-cnf-operator-macro-pipeline.yml
 - SVC-DEV/pipeline/nexus-cnf_operator/nexus-cnf-operator-issu-macro-pipeline.yml
 - SVC-DEV/pipeline/nexus-cnf_operator/nexus-cnf-operator-delete-macro-pipeline.yml